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A novel protocol for the efficient generation of all three major hippocampal neuronal sub-populations from human pluripotent stem cells

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Abstract

The diverse computational functions of the human hippocampus rely on coordinated interactions among dentate gyrus (DG), CA3, and CA1 subfields, yet generating all three neuronal identities in vitro - particularly CA1 - has remained challenging. Here we establish a reproducible and modular differentiation protocol that directs human pluripotent stem cells (hPSCs) through dorsomedial telencephalic progenitors to yield DG, CA3, and CA1 neuronal subtypes together with hippocampal regionally specified astrocytes. Early tri-inhibition combined with Sonic hedgehog suppression produced dorsal forebrain progenitors (FOXP1+, PAX6+), while FGF2 treatment supported progenitor maintenance and induced TBR2+ intermediate progenitors. Controlled WNT activation using CHIR99021 drove progressive enrichment of PROX1+ hippocampal progenitors across two independent donor lines. Terminal differentiation produced MAP2+/TAU+ neurons that expressed DG (PROX1), CA3 (GRIK4), and CA1 (WFS1, OCT6) markers, with maturing synaptic puncta. Defined progenitors generated long-lived (>400 days) hippocampal organoids exhibiting mixed neuronal-glial populations and spontaneous activity characterized by increased firing rates, high information entropy, and hub-like causal connectivity relative to monolayers, whereas astrocytes-supplemented monolayers displayed intermediate maturation. Population level electrophysiological analysis was also conducted to explore the dynamics of these different cultures. This platform enables systematic experimental control over neuron-astrocyte ratios, culture geometry, and developmental timing, providing a foundation for mechanistic studies of human hippocampal development, circuit function, and disease.

Note on figure quality

This is the preprint version of the manuscript. Figures are included adjacent to described results for the convenience of the reader but may be lower resolution due to file size restrictions on bioRxiv. High resolution figures are included as separate .tiff files for download.

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Introduction

The hippocampus functions as a dynamic computational hub for cognition, memory, and adaptive behaviour by integrating multimodal information and updating representations based on context and experience (1). Through interactions with cortical and subcortical regions, it supports relational and sequential memory, spatial mapping, and flexible inference, mediated by specialised circuits for pattern completion and unique networks of place and grid cells for spatial-temporal coding (1,1–3). These processes rely on balanced excitatory–inhibitory activity and plasticity mechanisms, including long-term potentiation (LTP) and long-term depression (LTD), which underpin learning and memory consolidation (4–6). Theoretical frameworks – from declarative and relational models to spatial navigation theories – highlight its central role in episodic and semantic memory and predictive modelling (1). Moreover, adult hippocampal neurogenesis is one of the few sources of new neurons in adult humans and has been recognised as a key contributor to brain plasticity and cognitive resilience (7,8). To understand these numerous essential functions, elucidating hippocampal cellular independent function, along with network interactions of the hippocampus within the wider brain, is essential.

In humans, the hippocampus sits as a C-shaped structure in the medial temporal lobe which originates from the dorsomedial telencephalon (9,10). It comprises the dentate gyrus (DG), Cornu Ammonis (CA1–CA4), and subiculum, which together form the trisynaptic circuit for memory processing. From the Cornu Ammonis, CA1 and CA3 have been identified as particularly critical for cognitive tasks (4,6,11–13). CA3 drives pattern completion and associative memory, CA1 integrates and outputs processed information, while the dentate gyrus and subiculum mediate cortical input and memory consolidation (14–17). The complexity of hippocampal function necessitates *in vitro* models to study its computational processes under controlled conditions. These models allow detailed investigation of spatial and temporal task performance, synaptic and network dynamics, and mechanisms underlying plasticity, memory formation, and disease. With the discovery of induced pluripotent stem cells (iPSCs), scalable and ethical generation of living region-specific neurons has become possible (18). Several protocols have been previously reported for generating iPSC-derived neurons based on a variety of different methods (19–23). Neural induction through the embryoid body (EB) formation is one of the most common approaches (19). Another widely implemented method for neural induction is inhibition of the transforming growth factor- β /SMAD signaling pathway by Noggin (LDN193189) and SB431542 (24). However, while these protocols often generate CA3 and DG neurons, CA1 neurons have been identified as a critical missing cell type from these methods (25,26).

To develop a differentiation protocol that was able to produce the full range of cell types of interest, we considered the ontogenetic development of the hippocampus. The hippocampus originates from the medial pallium structures of the dorsomedial telencephalon. The dorsomedial telencephalon consists of the choroid plexus, the

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cortical hem, and the medial pallium. The cortical hem lies between the choroid plexus and the medial pallium structures of the dorsomedial telencephalon. The cortical hem serves as a source of signaling cues secreting molecules, Wntless/Integrated (Wnt) and Bone Morphogenetic Protein (BMP). Wnt signaling, particularly via Wnt3a, plays a crucial role in initiating and organizing early hippocampal formation and neurogenesis through downstream transcription factors at the caudomedial margin of the continuous cerebral cortical neuroepithelium (27–30). In mice and humans, the role of Wnt3a in hippocampal specification has been highlighted (9,22). There are specific molecular markers that define hippocampal fields. The gene, GRIK4 (glutamate receptor, ionotropic, kainate 4) specifically the KA1 kainate receptor subunit, is enriched in CA3 (31), while Wolfram syndrome 1 (WFS1) and POU domain, class 3, transcription factor 1 (POU3F1, also known as OCT6) expression is enriched in CA1 (10,32). The gene, Prospero Homeobox 1 (Prox1) is also enriched in the DG. Finally, in hippocampal development and growth, Fibroblast growth factor (FGF) signaling has a multifactorial role-proliferation and survival (33,34).

Beyond only focusing on neurons however, hippocampal region-specific astrocytes were also considered a key target from the differentiation effort. Critical region-specific functional differences between astrocytes have also been observed, highlighting the need to be able to generate not just region-specific neurons, but also astrocytes (35–38). Astrocytes are the most abundant cell type in the brain, and their function was initially to support the neurons passively (39). However, current findings have found the role of astrocytes in the brain and their function in both neurodevelopmental and neurodegenerative diseases to be significantly more complex (40). As an example, astrocytes also play a key role in neuronal precursor guidance and maturation of neurons (41). Supporting this, coculturing of neurons and astrocytes substantially improves the functional maturation of human pluripotent stem cell-derived neurons (42). Alternatively, astrocytes and neurons can be differentiated from a common neural progenitor, giving rise to the two different cell types. However, the use of this method makes it difficult to monitor and control the exact percentage of astrocytes, and limits establishing different ratios of neurons to astrocytes, or even culturing only neurons as an experimental control (43,44). For this reason, we also explored adapting our protocol to be able to develop region-specific astrocytes from the hippocampal lineage.

Ultimately, this work aimed to develop and validate a new method for generating fully specified hippocampal differentiation, with a particular focus on CA1, CA3, and DG neurons, along with hippocampal region-specific astrocytes. This work was validated using immunocytochemistry and qPCR on hPSCs from two donors to capture both XX and XY chromosomal karyotypes. By using hPSCs from two donors and testing across both monolayers and organoids we were able to show the stability of this approach. Consequently, we also performed long-term electrophysiological recordings of each culture structure and tracked the differing maturation and properties of these cultures over time, including periods over a year for some.

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Results

Efficient generation of hippocampal progenitors from hPSCs

Hippocampal progenitors are known to originate from the medial pallium as directed by the signal molecules from the cortical hem (10). There are established differentiation protocols to induce neocortex, also known as cortical neuroepithelium (NE). Following a previously described neocortex induction method (**Figure 1A**), we induced hPSCs into dorsomedial telencephalic tissues using the tri-inhibition (dual SMAD and Wnt inhibition) protocol. This has been shown to induce Pax6+ cortical NE in culture (45) (**Figure 1B**). After 2 days of induction, we further blocked ventralisation by using cyclopamine, which inhibits the sonic hedgehog (Shh) signalling pathway (46,47). The differentiation robustly induced PAX6+ dorsal progenitors and OTX2+forebrain progenitor ($95 \pm 2\%$) with $98 \pm 1\%$ of these cells expressing the gene NESTIN, a marker for neural stem cells/progenitor cells (48) (**Figure 1B & C**). FOXG1 is a telencephalic marker, expressed from medial pallium to neocortex, and used to gauge the development of neuroprogenitors. We noted $96 \pm 0.5\%$ induction of FOXG1+ progenitors (**Figure 1B & C**). To investigate the expression of *EMX2* (49) and *GLI3* (50) expressed in the dorsomedial telencephalon, pro-neural gene, *PAX6* and *DCX* expressed in embryonic hippocampal progenitors (28,51), we analysed the RNA expression at day 11 in vitro (DIV11). Transcriptional analysis showed the increased levels of *EMX2*, *PAX6*, *DCX* and *GLI3* (**Figure 1D**). These observations, together with the immunocytochemistry data, demonstrate the induction of neural progenitors in the dorsomedial telencephalon region.

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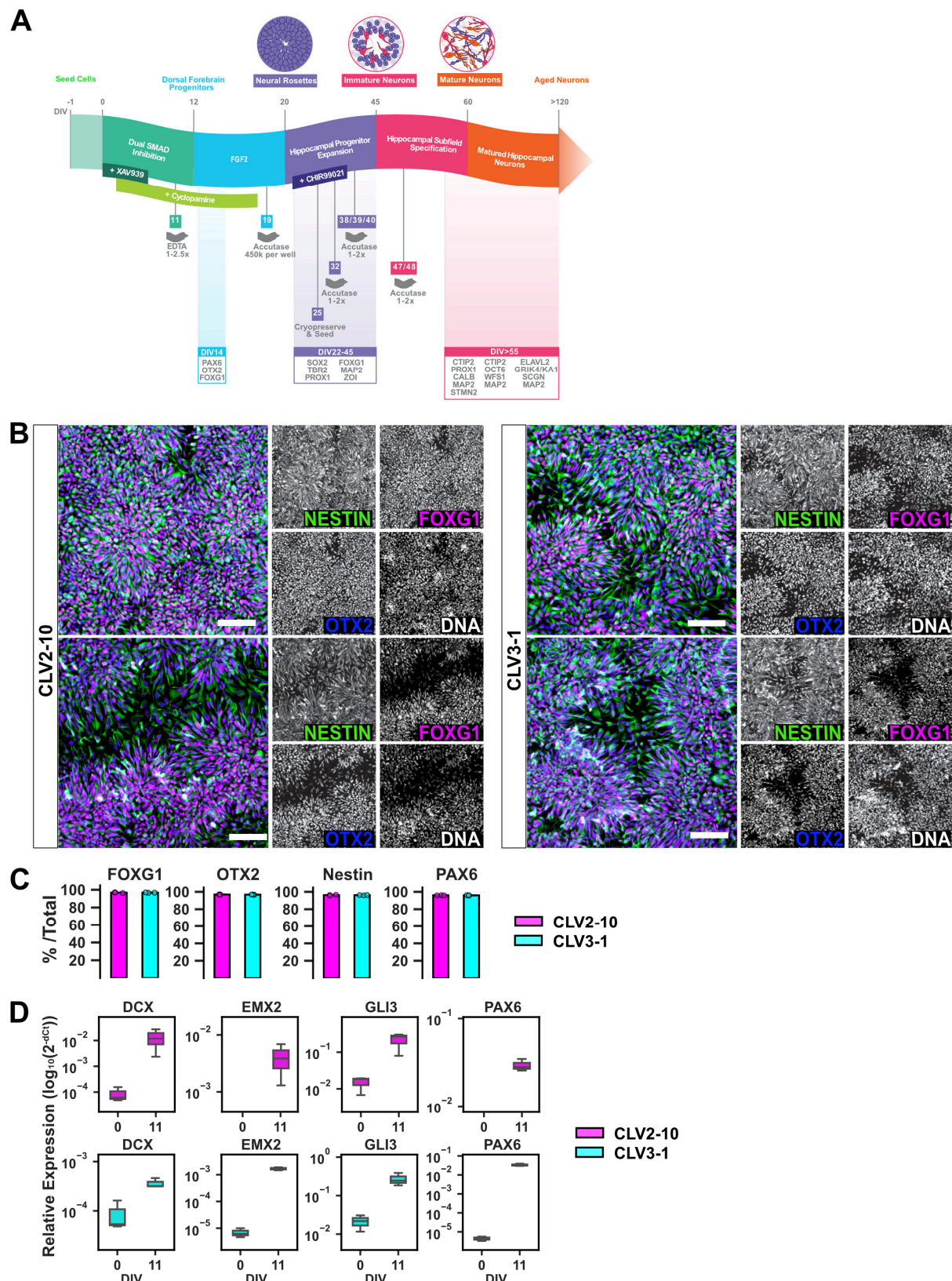


Figure 1: Efficient generation of dorsomedial telencephalic-like tissues.

A- Schematic representation of the differentiation protocol showing tri-inhibitory inhibition and SHH inhibition to generate dorsomedial telencephalic progenitors from hPSCs. Progenitors are further differentiated and matured till further analysis.

B- hPSCs-derived hippocampal progenitors at DIV14 express dorsomedial telencephalic markers OTX2, PAX6 and FOXG1 and the neural stem cell marker, NESTIN. Scale bar is 100 μ m.

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C- Quantification of *FOXG1*, *OTX2*, *PAX6* and *NESTIN* at DIV 14 for two hPSCs (male and female, $n = 3$ independent experiments each, data are mean $\pm$ SEM). The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

D- qRT-PCR confirms the expression of dorsomedial telencephalic markers. Boxplots with no data points indicate that no measurements were obtained for that marker on the given day in vitro.

We expanded the neural progenitors by treating the cultures with FGF2 for 8 days based on previously published findings (52). This promoted the generation of SOX2+ and FOXG1+ neural progenitor cells (NPCs) and TBR2 (also known as EOMES)-positive intermediate progenitor cells (IPCs). TBR2 is a transcription factor regulating both postnatal and adult hippocampal neurogenesis. In rodent models, TBR2 expression has been shown to modulate IPC proliferation and drive PROX1 expression (53). At DIV22, immunostaining confirmed the presence of TBR2+ IPCs ($5 \pm 1\%$) with the SOX2+ cells self-organizing into rosettes with ZO-1+ lumens (**Supplementary Figure 2A & B**). Transcriptional analysis of DIV 19-30 confirmed the increased expression of pro-neural genes (notably *PAX6*, *ASCL1*), embryonic hippocampal progenitor genes (notably *PROX1* and *DCX*), and downregulation of the pluripotency gene, *OCT4* (**Supplementary Figure 1A**).

WNT activation induces PROX1+ Hippocampal progenitors

Wnt signaling regulates proliferative or neurogenic fate decisions in hippocampal progenitors and has been shown to play a key role in hippocampal neurogenesis. Wnt signaling regulates PROX1+ expression, a marker essential for DG cell fate and adult hippocampal neurogenesis (23,54). The classic GSK-3 β inhibitor (Wnt agonist), CHIR99021, activates the canonical Wnt signaling pathway and upregulates the Wnt genes expressed in the cortical hem. Recent findings have shown that treating NPCs with CHIR99021 will induce neural progenitor cell proliferation and further differentiation of IPCs into PROX1+ cells (28,30). We next sought to investigate the impact of CHIR99021 treatment on number of cells as WNT signalling regulates cell proliferation and is essential in NPC proliferation in the medial pallium (30). We first tested 2 different concentrations of CHIR99021 (1.5 μ M and 3 μ M), and we noted the slight increase in number of nuclear positive (Hoechst+) cells in the 3 μ M-treated cells as compared to 1.5 μ M-treated groups. Immunostaining for PROX1 showed both conditions induced the differentiation of PROX1+ cells, with treatment of 3 μ M CHIR99021 leading to increased expression of PROX1+ and TBR2+ cells as compared to 1.5 μ M at DIV40. We then decided to go ahead with 3 μ M as previously published (9,55), and monitor the sturdy increase in PROX1 expression (**Supplementary Figure 1B**).

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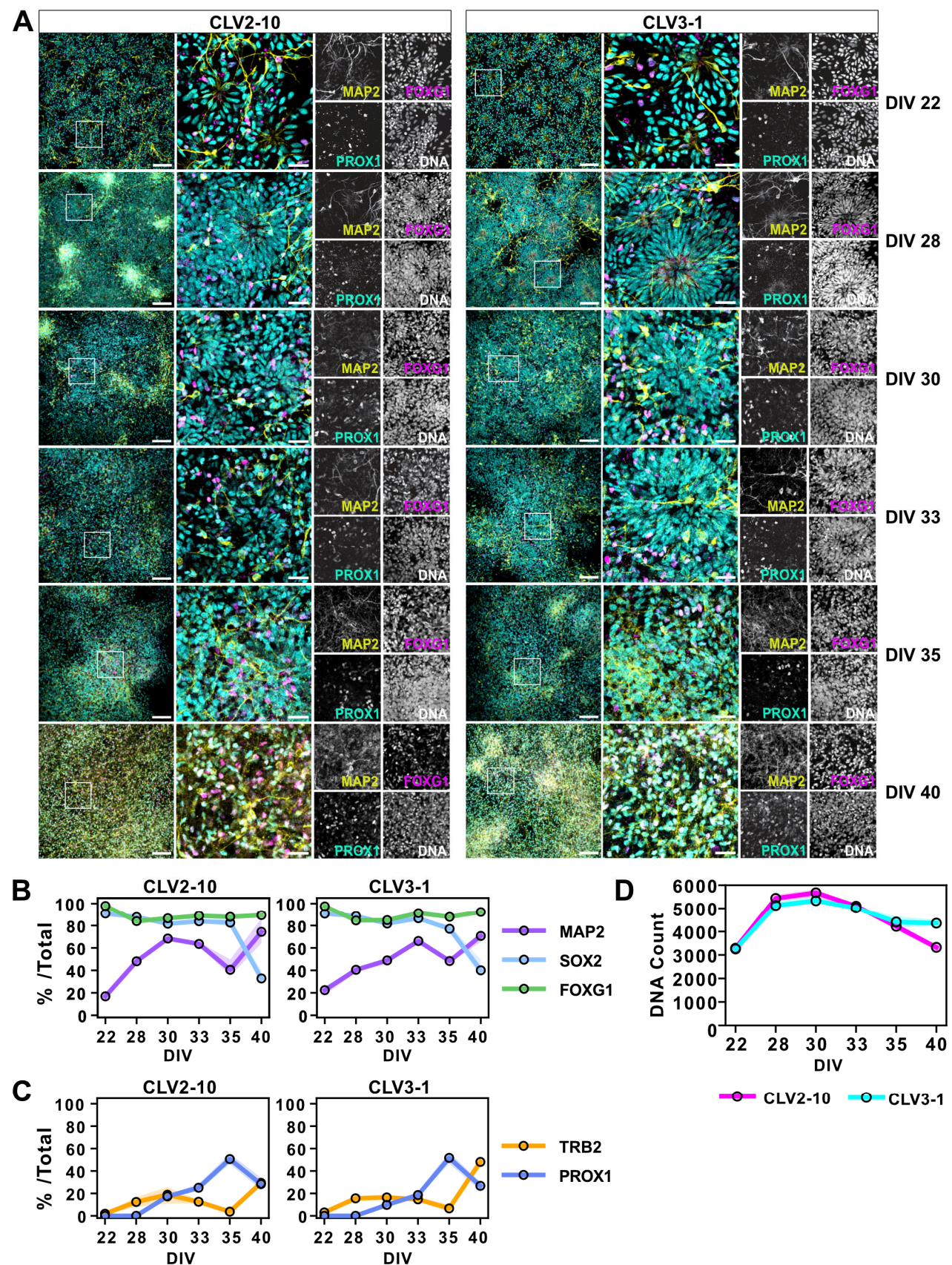


Figure 2: WNT activation induces TBR2+ IPCs and PROX1+ neurons.

A- Immunostaining of representative hPSCs-derived hippocampal progenitors at DIV 22 showing the appearance of PROX1+ cells from DIV22-40. Scale bar is 100 μ m and 25 μ m.

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B- Time-course quantification of FOXG1⁺ and SOX2⁺ progenitor populations and MAP2⁺ neuronal cells for two hPSC lines (male and female; $n = 3$ independent experiments each). Data are mean $\pm$ SEM, with confidence intervals defined by the minimum and maximum observed values. The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

C- Time-course quantification of TBR2⁺ IPCs and PROX1⁺ progenitor populations for two hPSC lines (male and female $n = 3$ independent experiments each, data are mean $\pm$ SEM). The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

D- CHIR99021 treatment increased total number of cells in culture for two hPSC lines (male and female $n = 3$ independent experiments each, data are mean $\pm$ SEM).

We found that PROX1 expression increased steadily from DIV 28-40. PROX1 expression was detected at DIV22, and its expression increased steadily after CHIR99021 treatment. As expected, the TBR2 and PROX1 expression increased steadily, which confirms previous findings in rodents (53). The expressions of FOXG1 and SOX2 remained unchanged over the days, suggesting the constant differentiation ability of the cells. The steady increase of both TBR2⁺ IPCs and PROX1⁺ cells indicates the proliferation and differentiation ability of the TBR2⁺ IPCs, giving rise to new PROX1⁺ and other neural progenitors (**Figure 2A, B & C**). Transcriptional analysis of cells at DIV 19-30 confirmed the induction of genes expressed in both hippocampal progenitors and differentiating cells (**Supplementary Figure 1A**). We confirmed the proliferative response to CHIR99021 by quantifying total cell numbers from DIV 22-40, during which cultures showed a ~1.5-fold increase before plateauing. These analyses were replicated across two independent cell lines, with no significant differences at any timepoint from DIV 22–40 ($p = 0.1$ – 1.0 , Mann–Whitney U test; details in Supplementary Table 3; **Figure 2D**). Taken together, these findings indicate that CHIR99021 enhances cell proliferation and supports the generation of hippocampal progenitors by increasing the production of PROX1⁺ neurons, with no detectable differences across the hPSC lines examined.

Generation of CA1 and CA3 hippocampal pyramidal- and granule-like neurons

To evaluate whether terminally differentiated neural progenitor cells give rise to hippocampal-specific neuronal subtypes, we analysed the expression of genes associated with the dentate gyrus (DG), CA1, and CA3 subfields (56). We first confirmed the increased expression of pro-neural genes (notably *PAX6*, *ASCL1*) and embryonic hippocampal progenitor genes (notably *PROX1* and *DCX*) by performing transcriptional analysis of DIV 37-54 (**Supplementary Figure 3**). Mature neurons are characterised by expression of microtubule-associated proteins (MAPs), which regulate cytoskeletal stability; TAU is the predominant MAP in mature neurons (57). In our cultures, MAP2 was used to label dendrites and TAU to identify axons and assess neuronal maturation. Immunostaining after DIV 60 confirmed the presence of $60 \pm 3\%$ MAP2⁺ cells (**Supplementary Figure 4A**).

During hippocampal development, the transcription factor BCL11B (also known as CTIP2) is expressed in postmitotic DG and CA1 neurons. Immunostaining for CTIP2

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confirmed the presence of postmitotic DG and CA1-like neurons, with about 20% of CTIP2-positive cells co-expressing MAP2. We also detected $12 \pm 5\%$ PROX1+ cells with about 50% of the population expressing MAP2+ confirming the presence of postmitotic DG neurons (**Supplementary Figure 4B**).

To further confirm and evaluate CA1 identity, we stained for WFS1, a marker of CA1 subfield neurons and the superficial marker, CALB (Calbindin) (58). The CALB marker shows a subtype-specific expression pattern, and it is present in glutamatergic neurons - including mature granule cells of the dentate gyrus (DG), superficial CA1 pyramidal neurons, and CA3 granule cells (59–62). Immunostaining revealed that $60 \pm 5\%$ of cells were WFS1+, with half of this population co-expressing CALB. We next confirmed the presence of SATB2+ neurons - a marker for CA1 hippocampal pyramidal neurons which plays a role in their differentiation, by immunostaining (63). Immunostaining confirmed the presence of $20 \pm 6\%$ SATB2+ cells (**Supplementary Figure 4A**).

In the human CA3 subfield, neurons are characterised by expression of ELAVL2 (HuB) and GRIK4 (KA1), which is expressed in a subset of CA3 hippocampus. At DIV 65, immunostaining confirmed $34 \pm 5\%$ ELAVL2+ cells with more than half co-expressing MAP2. Notably, we found ELVAL2+ cells co-expressing CTIP2 ($15 \pm 3\%$) (**Supplementary Figure 4B**), indicating an immature phenotype not yet committed to a specific hippocampal subfield.

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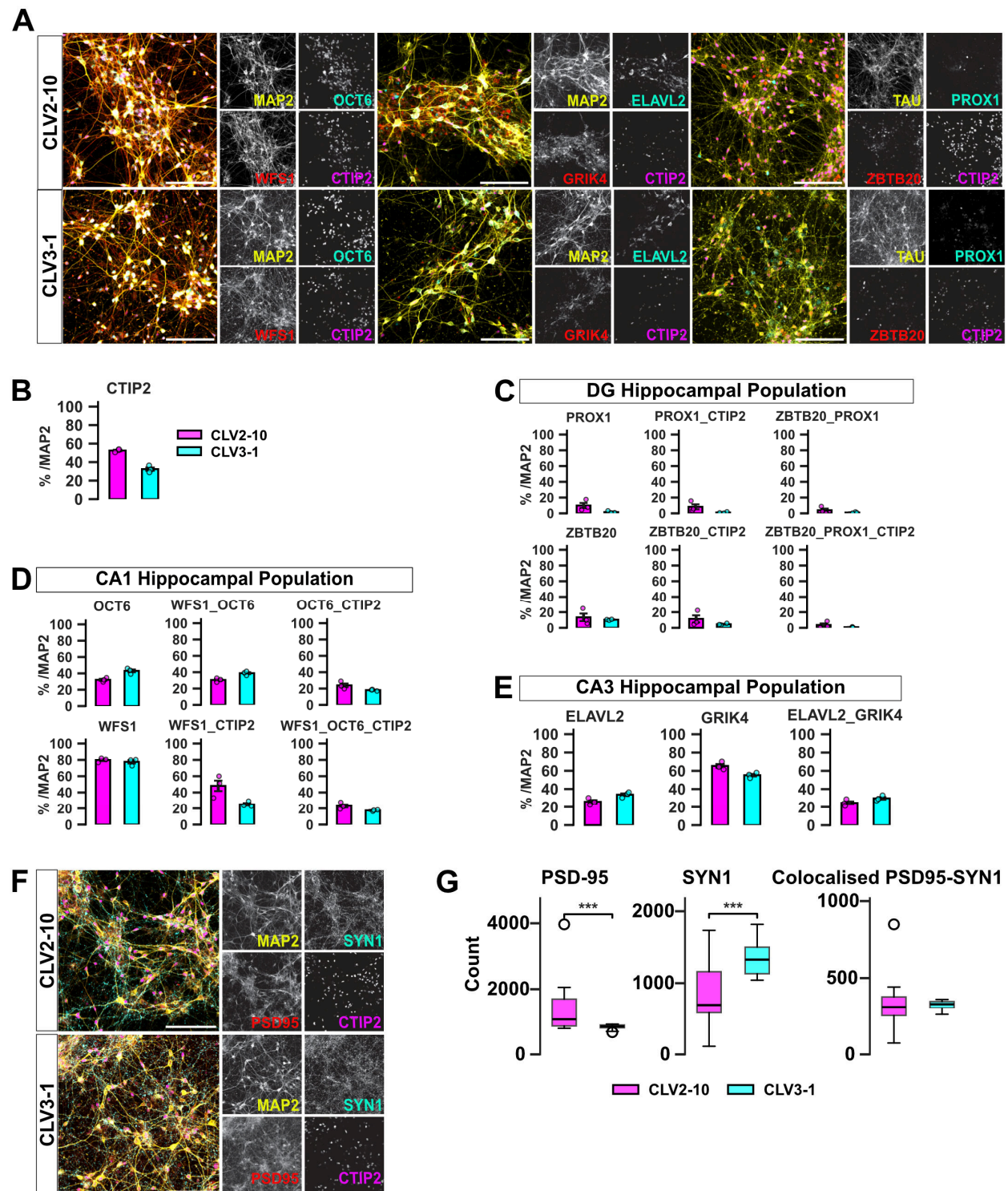


Figure 3: Generation of neurons constituting the different hippocampal subfields (DG, CA1 and CA3) upon long term culture (DIV 90).

A- Immunostaining analysis after long-term culture shows matured MAP2+ and TAU+ neurons constituting distinct hippocampal subfield fates: OCT6 (CA1), WFS1 (CA1), CTIP2 (CA1 and DG), PROX1 (DG) ZBTB20 (DG), ELAVL2 (CA3), GRIK4 (CA3).

B- Quantification of CTIP2+ cells in total MAP2+ cell population for two hPSC lines (male and female $n = 3$ independent experiments each, data are mean $\pm$ SEM).

C- Quantification of ZBTB20+, and PROX1+ cells in total MAP2+ cell population constituting distinct

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DG hippocampal neurons for two hPSC lines (male and female $n = 3$ independent experiments each, data are mean $\pm$ SEM).

D- Quantification of OCT6+ and WFS1+cells in total MAP2+ cell population constituting distinct CA1 hippocampal neurons for two hPSC lines (male and female $n = 3$ independent experiments each, data are mean $\pm$ SEM).

E- Quantification of ELAVL2+ and GRIK4+cells in total MAP2+ cell population constituting distinct CA3 hippocampal neurons.

F- Representative images of synaptic markers in hPSC-derived hippocampal neurons. Scale bar is 100 μ m.

G- Quantification of double-positive puncta, SYN1 and PSD95 colocalisation in MAP2+ cells for two hPSC lines (male and female, $n = 3$ independent experiments each, data are mean $\pm$ SEM).

To further characterise the maturation stage of differentiated neurons, cultures were maintained until DIV 90, followed by immunostaining for markers specific to DG, CA1, and CA3 hippocampal subfields (**Figure 3A**). Quantification focused exclusively on MAP2- and TAU-positive neurons to assess subfield-specific differentiation. More than 90% of the cells expressed MAP2 and TAU (**Figure 3A & Supplementary Figure 3C**). CTIP2 immunostaining revealed expression in $50 \pm 3\%$ of CLV2-10 cell line and $30 \pm 3\%$ of CLV3-1 cell line, accompanied by a reduced proportion of PROX1- and ZBTB20-positive neurons (**Figure 3B & 3C**). Notably, $75 \pm 3\%$ of WFS1-positive cells co-expressed MAP2, indicating neuronal identity.

To refine identification of CA1 neurons, OCT6 - a known CA1 marker - was used. Co-expression analysis showed that $30 \pm 3\%$ of cells were WFS1⁺OCT6⁺ across cell lines (**Figure 3D**). For CA3 characterisation, GRIK4 (KA1), a gene enriched in CA3 neurons, was used alongside ELAVL2. Immunostaining revealed that $24 \pm 3\%$ of cells were ELAVL2⁺GRIK4⁺ (**Figure 3E**). Given the known heterogeneity of CA3 subfield marker expression, and the predominance of ELAVL2⁺ cells lacking co-expression of CA3-enriched genes, we hypothesise that a subset of these cells may represent CA3 neurons at earlier stages of molecular maturation.

To assess the maturity of the cells, given that hPSC-derived hippocampal neurons are prone to slow maturation rates in vitro (64,65), we immunostained with pre-synaptic marker SYN1, and post-synaptic marker PSD-95, and observed an increased density of SYN1 puncta across both cells lines (**Figure 3F**). We used the Synbot open-source tool (66) to count the number of colocalized pre- and post-synaptic markers confirming the maturing state of the cultures (**Figure 3G**).

Morphologically, neurons resembling CA subfields exhibited large pyramidal somata, whereas DG-like neurons displayed smaller granule cell bodies, consistent with known hippocampal cytoarchitecture. We identified neurons which were positive for ELAVL2 and WFS1 with distinct large pyramidal somata appearing in the cultures indicating the maturation phase of the culture. Most of the neurons possessed immature dendrites with shorter processes and fewer branches (**Supplementary Figure 3C**).

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Generation of functional hippocampal organoids

There are reports on generating hippocampal organoids (HO) from hPSCs using embryoid bodies. The asynchronous differentiation, which leads to the presence of choroid plexus tissues, results in smaller organoids over extended culture periods, with some HO unable to survive longer than 100 days in culture (9,23). We generated the HOs with fully committed hippocampal progenitors after confirming their identity in 2D cultures (see **Figure 2A, B, C & D**). The HOs maintained their spherical structures throughout the growth period, measuring between 750-1000 μm in diameter (**Figure 4A**). The expression of FOXG1, PROX1 and CTIP2 at DIV40 indicated the presence of hippocampal progenitors and DG neurons with more than 50% expressing MAP2 and CALB across both cell lines (**Figure 4B & C**). The presence of CALB expression further confirms the different specific subtypes of neurons in the HO (**Figure 4B**). The HOs did not develop any cavities after being cultured for over 400 days. This confirms that the differentiation protocol restricts the presence of choroid plexus epithelial cells (23). Immunostaining at DIV 200 confirmed the presence of PROX1+ DG neurons and WFS1+ CA1 neurons across both cell lines (**Figure 4D**). This further demonstrated the successful induction of organoids into a hippocampal fate with no significant difference between cell lines.

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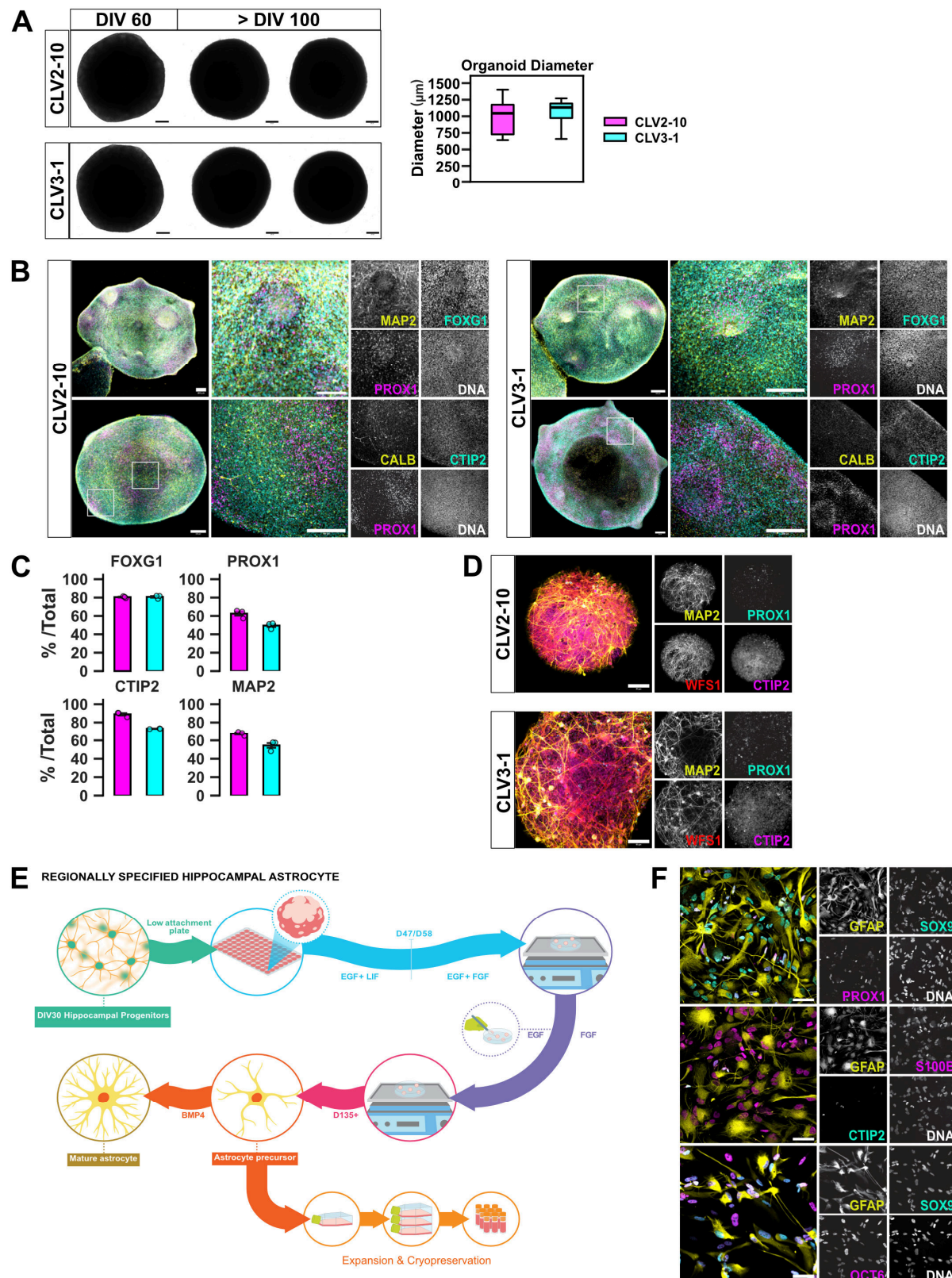


Figure 4: Generation of region-specific hippocampal organoids and astrocytes.

A- Morphology of generated HOs at two different ages, DIV60 and DIV100 (left panel). Quantification of the diameter of the organoids (right panel; both stages are combined). Scale bar is 200 μm.

B- Immunostaining analysis of DIV 40 HOs to confirm the presence of FOXG1+ progenitors and PROX1+ neurons for two hPSC lines (male and female, n = 3 independent experiments each). Scale bar is 200 μm.

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C- Quantification of FOXG1+, CTIP2+, PROX1+ and MAP2+ cell population at DIV 40 in the HO. The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

D- Immunostaining analysis of HO at DIV200 showing CA1 and DG cell population. Scale bar is 50 μ m.

E- Schematic representation of the differentiation protocol to generate hPSC-derived regionally specified hippocampal astrocytes.

F- Immunostaining analysis of hPSC-derived regionally specified hippocampal astrocytes showing GFAP+, S100 β + and SOX9+ astrocytic markers and OCT6+, CTIP2+ and PROX1+ hippocampal markers. Scale bar is 50 μ m.

There are reports on generating hippocampal organoids (HO) from hPSCs using embryoid bodies. The asynchronous differentiation, which leads to the presence of choroid plexus tissues, results in smaller organoids over extended culture periods, with some HO unable to survive longer than 100 days in culture (9,23). We generated the HOs with fully committed hippocampal progenitors after confirming their identity in 2D cultures (see **Figure 2A, B, C & D**). The HOs maintained their spherical structures throughout the growth period, measuring between 750-1000 μ m in diameter (**Figure 4A**). The expression of FOXG1, PROX1 and CTIP2 at DIV40 indicated the presence of hippocampal progenitors and DG neurons with more than 50% expressing MAP2 and CALB across both cell lines (**Figure 4B & C**). The presence of CALB expression further confirms the different specific subtypes of neurons in the HO (**Figure 4B**). The HOs did not develop any cavities after being cultured for over 400 days. This confirms that the differentiation protocol restricts the presence of choroid plexus epithelial cells (23). Immunostaining at DIV 200 confirmed the presence of PROX1+ DG neurons and WFS1+ CA1 neurons across both cell lines (**Figure 4D**). This further demonstrated the successful induction of organoids into a hippocampal fate with no significant difference between cell lines.

Brain organoids can generate functional glial populations, including astrocytes and oligodendrocytes (23,67). To assess this capacity in our HOs, we immunostained with astrocytic markers GFAP and S100 β . We detected the presence of astrocytes at DIV 90 that increased progressively, reaching proportions comparable to neurons by DIV 400 (**Supplementary Figure 5A & B**).

Generation of hippocampal regionally specified astrocytes

Astrocytes, distinguished by their highly ramified morphology, contribute to brain function and behaviour through homeostatic maintenance and synaptic regulation. They also play an active role in modulating neuronal network activity. These roles are mediated by their close physical proximity to neurons and dynamic interactions within neural circuits (36). To generate regionally specified neurons, we followed previously established protocols. Hippocampal neural progenitor cells (NPCs) were aggregated into spheres at DIV 19 and treated with CHIR99021 from DIV 22 until DIV35. Cultures were maintained until DIV35, at which point over 30% of cells expressed the hippocampal marker PROX1 (refer to **Figure 2D**). To induce astrocytic

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lineage commitment, the medium was switched to astrocyte differentiation conditions, supplemented with human leukemia inhibitory factor (hLIF) and epidermal growth factor (EGF) for four weeks. Subsequently, cultures were maintained in medium containing EGF and fibroblast growth factor (FGF) for an additional 2-3 months (**Figure 4E**).

Following dissociation and plating, the cultures predominantly contained non-neuronal cells with few neuron-like morphologies (**Figure 4F**). Immunocytochemical analysis revealed that the hPSC-derived astrocytes expressed canonical astrocytic markers, including glial fibrillary acidic protein (GFAP), S100 calcium-binding protein B (S100 β), and SOX9, while lacking expression of the neuronal dendritic marker MAP2. To assess regional identity, cells were further immunostained for hippocampal markers CTIP2, PROX1, and OCT6. A small population of CTIP2-positive cells with reduced nuclear size was observed. Additionally, cells co-expressing SOX9, S100 β , and GFAP also exhibited expression of OCT6, CTIP2, and PROX1, supporting the regional specification of the astrocyte population (**Figure 4F**).

Functional analysis of dissociated hippocampal-like neurons, hippocampal neurons with astrocytes and hippocampal organoids using electrophysiology

To assess neuronal functionality and network integration, multi-electrode array (MEA) recordings were performed across three hippocampal culture groups (**Figure 5A**). Hippocampal differentiation was performed either as a monolayer culture of hippocampal neurons (HN) or as a three-dimensional hippocampal organoid (HO). In addition, hippocampal neuron cultures supplemented with astrocytes at a standard 1.4:1 neuron-to-astrocyte ratio were prepared (HN+HA). Periodic recordings captured spontaneous activity patterns, enabling quantitative comparison of firing dynamics and network organization across conditions.

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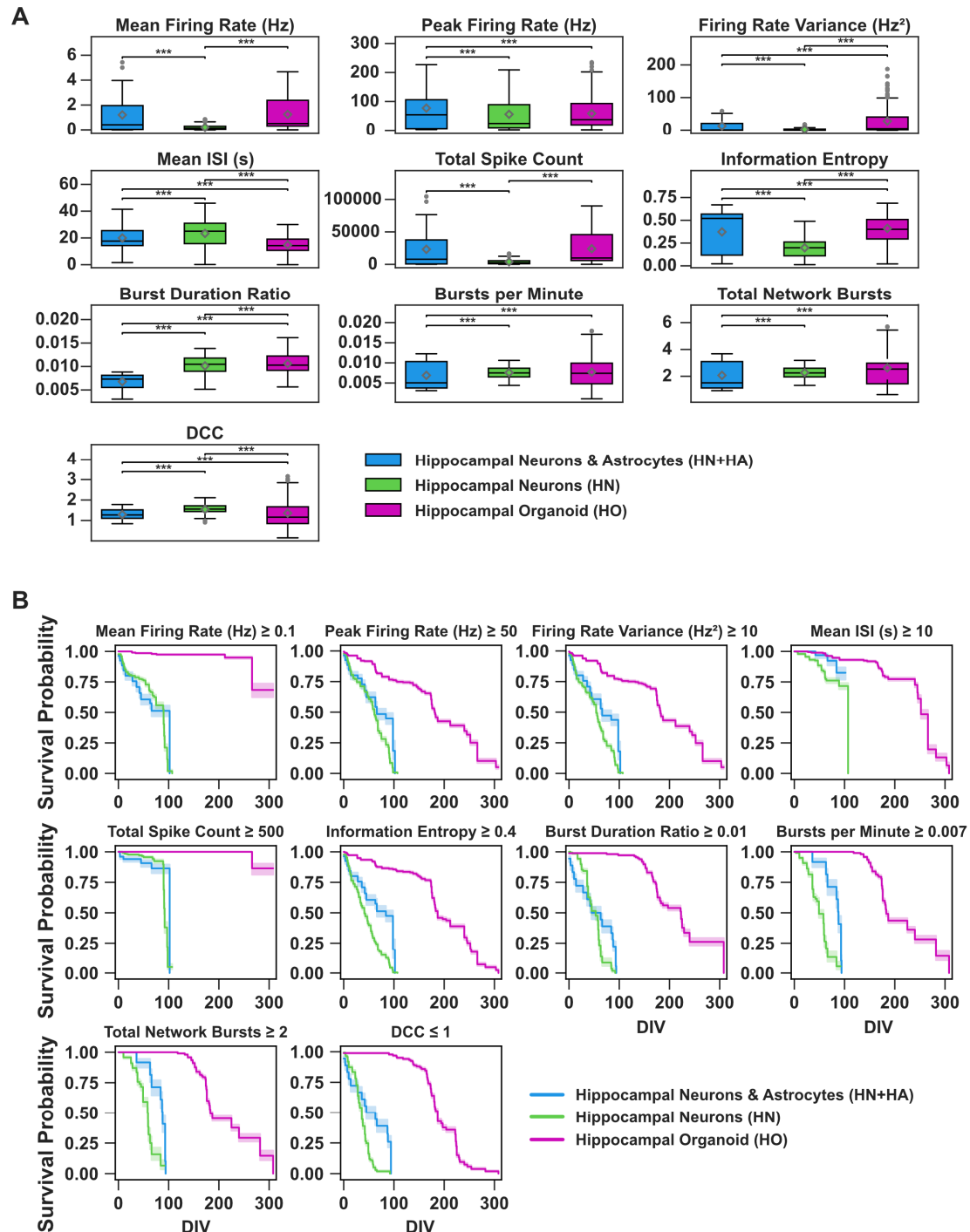


Figure 5: Functional maturation and longevity differ across different hippocampal culture systems.

A- Distribution of electrophysiological and network-level metrics across three hippocampal culture groups: HN+HA (blue), HN (green), and HO (magenta). Metrics include mean firing rate, peak firing rate, firing rate variance, mean inter-spike interval (ISI), total spike count, information entropy, burst duration ratio, bursts per minute, total network bursts, and deviation from criticality (DCC). Boxplots show the median (center line), interquartile range (box), and $1.5 \times$ IQR (whiskers); individual points represent cultures. Diamonds denote group means. Statistical comparisons between groups were performed using pairwise tests with multiple-comparison correction; significance levels are indicated

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(Games–Howell post hoc tests; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; see Supplementary Tables 24 and 25).

B- Kaplan–Meier survival curves showing culture longevity as a function of meeting predefined functional activity thresholds for each metric shown in (A). Survival probability is plotted against days in vitro (DIV) for each culture group. Thresholds were chosen to reflect minimal functional or network-level activity (e.g., firing rate, bursting, entropy, or criticality criteria). Cultures failing to meet the specified threshold were considered to have failed the criterion. Shaded regions indicate 95% confidence intervals. Across nearly all metrics, HO cultures exhibit significantly prolonged survival and sustained functional activity compared to HN or HN+HA cultures (log-rank tests; see Supplementary Table 26).

The electrophysiological data revealed marked differences in spontaneous activity, reflecting distinct stages of neuronal maturation and network formation (**Figure 5A**). The HO group exhibited the highest overall excitability, characterized by significantly elevated mean and peak firing rates, increased firing rate variance, and higher total spike counts (Games–Howell post hoc tests; p -values correspond to mean firing rate, peak firing rate, firing rate variance, and total spike count, respectively: HO vs HN, $p < 0.001$, 0.970 , < 0.001 , < 0.001 ; HO vs HN+HA, $p = 0.001$, 8.6×10^{-12} , < 0.001 , < 0.001 ; see Supplementary Tables 24 and 25). These cultures also showed longer burst durations, higher burst rates, and greater numbers of network bursts, indicating the emergence of highly synchronized and recurrent network activity consistent with mature functional connectivity.

In contrast, astrocyte-supported monolayers (HN+HA) displayed intermittent but structured firing, with moderate firing rates, reduced variability, and intermediate burst metrics (**Figure 5A**). This activity profile suggests partial yet organized network formation, likely facilitated by astrocytic support of synaptic development and homeostatic regulation. Meanwhile, the HN group exhibited minimal spontaneous activity, with low firing rates, and sparse bursting, consistent with immature or weakly interconnected networks (**Figure 5A**).

To further assess network stability and information-processing capacity, we quantified each culture's deviation from criticality coefficient (DCC) and mean information entropy (**Figure 5A**). Across all groups, DCC values indicated that spontaneous population dynamics were positioned away from the critical regime. Previous work has shown that exposure to structured, task-positive environments can drive in vitro neuronal networks toward dynamical states closer to criticality, where information transmission and adaptive behaviour are maximised (56). Within the present spontaneous setting, HO exhibited the lowest DCC values, indicating dynamics comparatively closer to criticality. In contrast, the HN group showed significantly larger deviations from criticality (Games–Howell post hoc tests: HN vs HO, $p < 0.001$; HN vs HN+HA, $p < 0.001$; see Supplementary Table 25), consistent with more subcritical dynamics, while the astrocyte-supported monolayers displayed intermediate DCC values.

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Parallel trends were observed in information entropy, with HOs exhibiting significantly higher mean entropy (Games–Howell post hoc tests: HO vs HN, $p < 0.001$; HO vs HN+HA, $p = 1.8 \times 10^{-5}$; see Supplementary Table 25), indicative of rich and diverse firing patterns, whereas HNs showed the lowest entropy, reflecting reduced variability and limited informational complexity. The HN+HA group again occupied an intermediate regime, consistent with their moderated yet structured activity patterns.

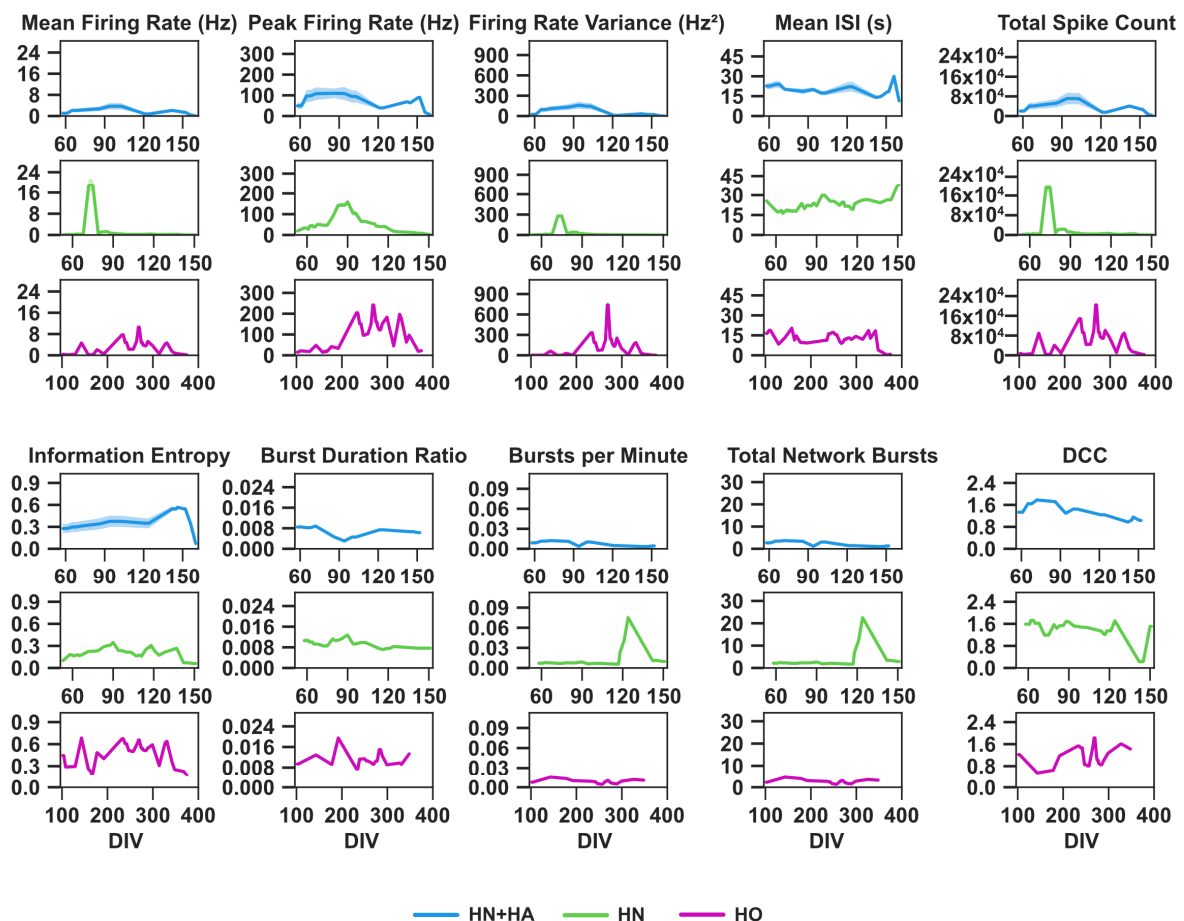


Figure 6: Longitudinal evolution of electrophysiological and network dynamics across hippocampal culture systems.

Time-resolved trajectories of electrophysiological and network-level metrics plotted as a function of days in vitro (DIV) for three hippocampal culture groups: HN+HA (blue), HN (green), and HO (magenta). Each trace represents the population mean across cultures within a group, computed using a 7-day moving average, with shaded regions indicating 95% confidence intervals. Note the extended recording duration for HO cultures relative to monolayer cultures, reflecting their prolonged functional stability. These longitudinal profiles reveal distinct maturation trajectories and regimes of network activity across culture systems, with organoids exhibiting sustained, heterogeneous dynamics over extended timescales.

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Survival analyses further demonstrated that the HO group maintained functional electrophysiological thresholds across multiple metrics for significantly longer relative time periods compared to the other groups (log-rank tests; see Supplementary Table 26), while HN cultures rapidly dropped below these thresholds (**Figure 5B**). HN+HA cultures showed intermediate stability closer to the HN group, reinforcing a graded progression from immature to highly integrated network states across the three culture conditions.

Together, these results demonstrate progressive functional maturation across the three models, with organoid cultures showing the most robust and integrated electrophysiological signatures, astrocyte-supported monolayers displaying intermediate functionality, and pure neuronal monolayers remaining largely immature.

To examine how spontaneous electrophysiological activity evolved over time, we computed 7-day moving averages of firing and bursting metrics across days in vitro (DIV), providing a view of developmental trajectories and neuronal maturation (**Figure 6**). Across all culture conditions, firing-related measures exhibited structured, non-monotonic trajectories, with early increases followed by stabilization or partial decline, consistent with progressive maturation and homeostatic regulation of excitability. HO cultures displayed a prolonged and heterogeneous developmental profile, characterized by sustained elevations and fluctuations in mean and peak firing rates, firing rate variance, total spike counts, and information entropy over extended time periods, reflecting slower yet more persistent maturation dynamics. In contrast, HN+HA and HN cultures reached more stable activity regimes earlier, with lower overall firing rates, bursting measures, and entropy peaking at mid-DIV and subsequently plateauing or modestly decreasing. Notably, HN+HA cultures exhibited smoother temporal trajectories and reduced variability compared to HN cultures, suggesting a stabilizing influence of astrocytes on developing neuronal excitability. DCC showed distinct temporal trajectories across culture types, with HO cultures exhibiting an increase over DIV, despite maintaining lower median DCC values than HN cultures when considered in aggregate, indicating that organoids undergo a developmental shift in criticality-related dynamics during spontaneous activity, in contrast to the earlier stabilisation observed in monolayer cultures (**Figure 6 & Figure 5A**). Together, these temporal profiles highlight distinct maturation kinetics across culture models, with organoids exhibiting extended developmental windows and more sustained electrophysiological activity.

Causal Connectivity and Network Topology

Next, directed functional interactions among electrodes were examined using Granger causality-based connectivity graphs (see Supplementary Methods for details). This approach quantifies the extent to which the activity of one electrode

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can predict that of another, allowing inference of directed, weighted network architectures underlying spontaneous activity.

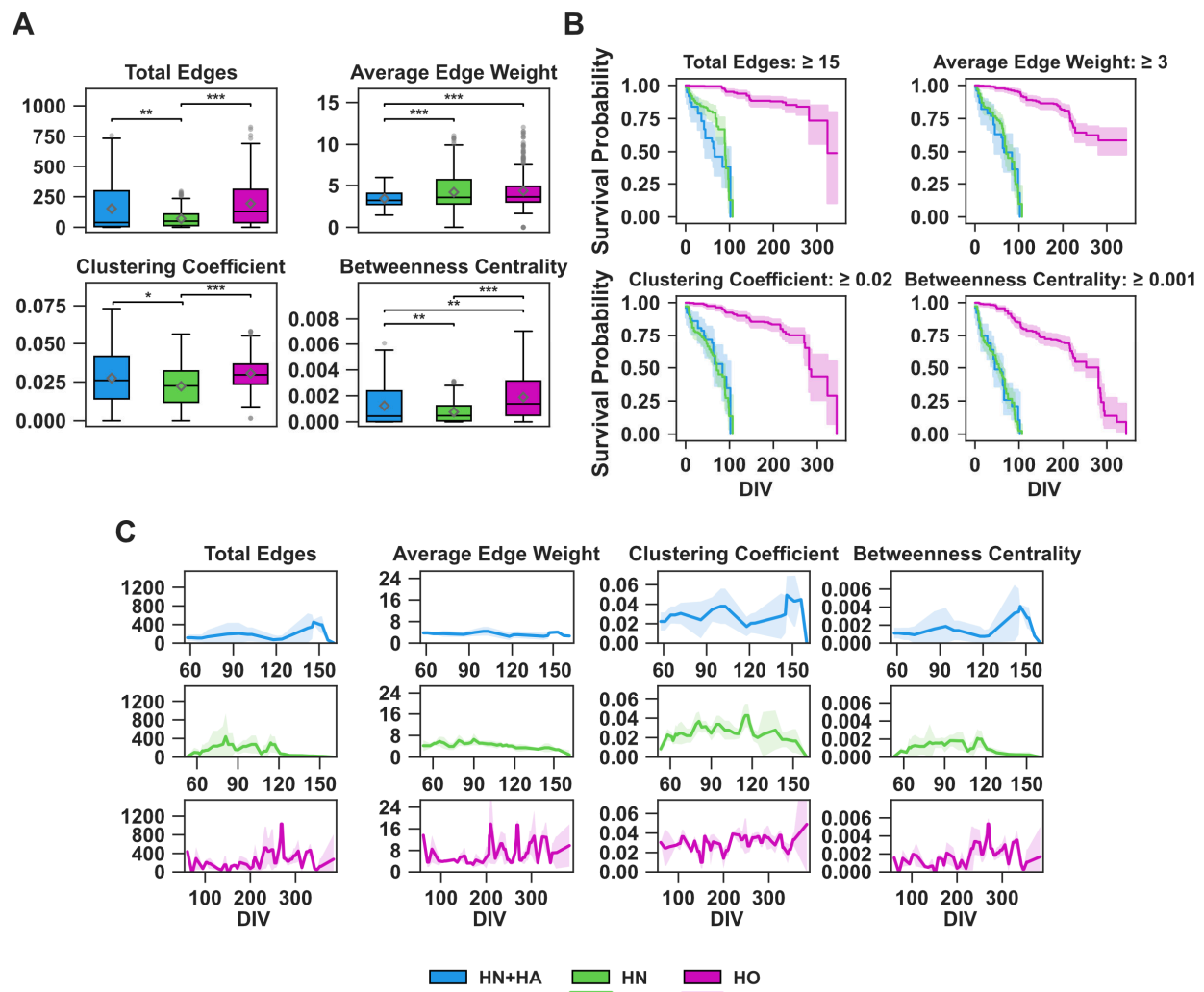


Figure 7: Organization and stability of hippocampal network connectivity across culture systems.

A- Distribution of connectivity metrics computed from causal connectivity networks across HN+HA (blue), HN (green), and HO (magenta) cultures. Metrics include total number of edges, average edge weight, clustering coefficient, and betweenness centrality. Boxplots indicate the median (center line), interquartile range (box), and $1.5 \times$ IQR (whiskers), and diamonds indicate group means. Statistical comparisons between groups were performed using pairwise tests with multiple-comparison correction; significance levels are indicated (Games–Howell post hoc tests ; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; see Supplementary Tables 27 and 28).

B- Survival probability curves showing the fraction of cultures that maintain connectivity-related metrics above predefined thresholds as a function of days in vitro (DIV). Thresholds were applied to total edges, average edge weight, clustering coefficient, and betweenness centrality (values indicated in each panel). Cultures were considered to have failed a criterion at the first time point at which the metric dropped below the threshold. Shaded regions indicate 95% confidence intervals. Across all connectivity criteria, HO cultures exhibit significantly prolonged maintenance of network structure relative to monolayer hippocampal cultures (log-rank tests; see Supplementary Table 29).

C- Longitudinal evolution of causal connectivity metrics plotted as a function of DIV for each culture group. Traces represent group means at each time point, computed using a 7-day moving average,

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with shaded regions indicating 95% confidence intervals. HO cultures display sustained and heterogeneous network connectivity over extended timescales, whereas monolayer cultures exhibit earlier declines in network complexity and integration.

Across culture conditions, Granger causality–derived networks exhibited marked differences in topology, connectivity strength, and temporal stability (**Figure 7A**). Aggregated network statistics revealed that HO cultures formed the most densely connected and heterogeneous networks, with significantly higher total edge counts and strong average edge weights (Games–Howell post hoc tests; p -values correspond to total edge count and average edge weight, respectively: HO vs HN, $p < 0.001$, $p = 0.388$; HO vs HN+HA, $p < 0.001$, $p < 0.001$; see Supplementary Tables 27 and 28). These networks also displayed elevated clustering coefficients and significantly higher betweenness centrality (Games–Howell post hoc tests; p -values correspond to clustering coefficient and betweenness centrality, respectively: HO vs HN, $p < 0.001$, $p < 0.001$; HO vs HN+HA, $p = 0.178$, $p < 0.001$; see Supplementary Table 28), indicating the presence of locally clustered structure alongside prominent hub-like nodes that mediate information flow across the network. In contrast, HN cultures exhibited sparse connectivity, with lower total edge counts, and reduced centrality measures, consistent with weakly integrated network architectures. The HN+HA group occupied an intermediate regime, showing greater connectivity and clustering than HN cultures but less pronounced network integration than the HO group.

To assess the temporal persistence of network organization, we performed survival analyses based on thresholded network metrics (**Figure 7B**). HO cultures maintained high levels of connectivity (i.e. total edges), average edge strength, clustering, and centrality for significantly longer relative time periods compared to both monolayer conditions (log-rank tests; p -values correspond to total edge count, average edge weight, clustering coefficient, and centrality, respectively: HO vs HN, all $p < 0.001$; HO vs HN+HA, all $p < 0.001$; see Supplementary Table 29), indicating sustained preservation of complex network structure. In contrast, both monolayer groups rapidly fell below these thresholds, reflecting transient or unstable network organization. Finally, analysis of 7-day moving averages across days in vitro revealed distinct developmental trajectories in network topology (**Figure 7C**). Organoid networks exhibited prolonged and dynamic reconfiguration, with gradual increases and sustained variability in edge counts, clustering, and centrality measures over extended developmental windows. Monolayer cultures, by comparison, showed earlier stabilisation of network properties, with HN+HA cultures maintaining more consistent connection profiles than HN cultures. Together, these results demonstrate that three-dimensional organoid architectures support the emergence, persistence, and continued reorganization of complex directed causal and functional networks, whereas two-dimensional monolayers converge more rapidly toward simpler and more stable connectivity regimes.

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Discussion

In this study, we have demonstrated a robust and modular approach for the generation of dorsomedial telencephalic region progenitors from hPSCs through a modified tri-inhibitory induction method (45,68). Introducing the FGF-MAPK signalling by treating the cells with FGF2 led to the maintenance of the progenitor phenotype and the expression of TBR2+ IPCs, which play a role in hippocampal progenitor fate specification. This study established a protocol which demonstrates the increased expression of PROX1+ cells in cultures. When cultured over time, these progenitors gave rise to all key hippocampal subtypes, including dentate gyrus (DG), CA3 neurons, and – until now – the elusive CA1 neuron subtypes. While earlier protocols enabled DG and CA3 neural differentiation, until this work, a reliable protocol that generates a relatively meaningful number of neurons with a CA1 identity has been notably absent (9,19,25,26). Given the known critical role of the CA1 neural subtype in integrating and outputting processed information, this has been a significant limitation in understanding the dynamics of hippocampal cells that can now be overcome (5,6,12,13). Moreover, it was established that the range of subtypes is not a feature of the donor line, as the results were replicated with two independent hPSC lines from different donors. It is proposed that future research will be able to use the protocol described in this work to explore and better understand the computational properties of hippocampal neural cultures with a range of subtypes (69,70).

Mechanistically, the staged manipulation of signalling pathways appears central for the protocol to guide early dorsomedial identity toward hippocampal lineages. Early tri-inhibition, combined with Shh pathway inhibition to prevent ventralisation (46,47), produced high proportions of dorsal forebrain progenitors, consistent with a dorsomedial telencephalic starting point. Subsequent exposure to FGF2 was associated with maintenance of a progenitor phenotype and emergence of TBR2+ intermediate progenitor cells, a lineage state implicated in hippocampal neurogenesis and the progression toward PROX1+ DG programs in vivo. This intermediate progenitor phase likely creates a permissive developmental window in which WNT activation (via CHIR99021) can drive hippocampal progenitor programs, including sustained increases in PROX1 expression over time (33,34). Notably, the observed sensitivity to CHIR99021 dose is consistent with WNT's dual role in regulating proliferation and fate decisions (30). These results emphasise that controlled WNT activation can be used in a tuneable manner, as a morphogen that shapes progenitor expansion and downstream differentiation trajectories.

Along with differentiation standard monolayers, organoids were also generated, as were hippocampal regionally specified astrocytes to allow the astrocyte proportion of the cultures to be further enhanced to more physiologically relevant levels of expression. These neural cultures survived and were electrically active for extended periods of time, in the case of the organoids, over 400 days. This supports the idea that these are a useful model for long-term assessment for neurodevelopment, toxicology and pharmacology, and for studying neural dynamics. These cultures

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were electrically active networks, with functional synaptic connections identified, as confirmed by electrophysiological activity and immunocytochemical analysis. Of particular interest was the long-term and detailed comparison of nuanced electrophysiological activity across the different culture types. The firing rate data showed that while organoids persisted in electrophysiological activity for longer and did have significantly higher mean firing rates than monolayers, peak firing rates did not follow the same pattern. The organoids also showed closer to critical dynamics compared to the monolayer cultures initially, although these further deviated as time went on. This finding is consistent with recent work showing that organoids can maintain closer to critical dynamics than monolayers, but that structured stimulation is required for a cell culture to maintain itself in a critical state (71,72). Overall while monolayers with additional astrocytes did seem to partially enhance electrophysiological activity, these results suggest that full maturation and network integration are better achieved in the organoid model compared to traditional monolayers. Another notable finding is that amongst the differences in the functional connectivity of the various cultures, the betweenness centrality (a measure of how many required connectors exist in a network) was significantly higher in organoids compared to both monolayer cultures.

That region-specific astrocytes could be reliably generated with a small variation on the original protocol is also noteworthy. A growing body of literature has identified that astrocyte function is region-specific, with key differences in how these glial cells interact and support neurons (35–38). For future work that aims to understand the functional interactions between astrocytes and neurons, along with how different neuron-astrocyte ratios may influence network dynamics, the ability to more tightly control the proportions of these cells in a region-specific manner will be valuable. While a detailed analysis of these aspects was beyond the scope of this study which focused on the differentiation methods and validation, several key electrophysiological differences between standard monolayers and those supplemented with additional astrocytes were observed. Key differences included the astrocyte-supplemented monolayers having a higher mean firing rate, shorter bursts, and increased information entropy levels than standard hippocampal monolayers. The mechanisms behind these functional differences will be of key importance to uncover in future research.

A key limitation of this work however is that in terms of cell culture structure, it only compared organoids to monolayers. As such this work cannot conclusively answer whether these electrophysiological properties are a result of the organoids having a specific spherical morphology, having greater cell density, or some other feature. Future work will be well suited to explore in greater detail the structure-function relationship with these cell types, including the use of modular cell types that specifically control geometry and topology of these cultures (73). Another limitation is that although marker-based immunostaining supports the presence of DG, CA1, and CA3 neurons, the observed overlap in some marker combinations and the persistence of immature dendritic morphology indicate that a subset of cells may

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remain in transitional states. It is also possible that the cellular identity may not yet be fully resolved for all neurons even at later timepoints. Single-cell transcriptomic profiling across differentiation stages, combined with trajectory inference anchored to human fetal hippocampal datasets, would clarify lineage progression and help refine conditions that bias maturation toward more stable subfield-associated programs. Second, while MEA recordings demonstrate network activity and maturation differences across culture formats, they do not directly resolve synapse type (excitatory versus inhibitory), cell-type-specific contributions, or trisynaptic-like circuit motifs. Incorporating targeted interneuron differentiation, more explicit excitatory/inhibitory balancing, and stimulation paradigms (including patterned theta-like stimulation) would strengthen links between the observed network behaviours and how the hippocampal functions are central to learning and memory.

Ultimately, this work has demonstrated an improved protocol for generating all major hippocampal subtypes from hPSCs. Not only does this approach generate the long-elusive CA1 subtype that has been missing from previous protocols, but the method is able to produce region-specific astrocytes. The rich electrophysiological properties of the neural cultures were found to be sensitive to morphology (monolayer vs organoid) and astrocyte to neuron ratio. While significant work remains to elucidate the complex workings of the hippocampus, the individual cell types, and understand inter-region interactions, the ability to culture a more diverse array of neural subtypes will be an invaluable tool for future research.

Methods

hPSC maintenance and differentiation

hPSC lines CLV2.10 and CLV3.1 were maintained as previously described(74). Hippocampal induction was achieved using an established cortical induction protocol with adaptations (45,75). hPSCs were seeded into Laminin-521-coated (5 µg/mL) plates at $0.15 \times 10^6/\text{cm}^2$ in mTeSR1 medium with 50 nM Chroman 1 (MedChem Express). Twenty-four hours later, cells were switched to dual-SMAD and Wnt inhibition for 6 days in Basal differentiation medium (BDM) supplemented with 100 nM LDN193189 (Stemgent), 10 µM SB431542 (R&D Systems) and 1µM XAV-939. Basal differentiation medium consisted of 1:1 DMEM/F12 and Neurobasal with 0.5× B27, 0.5× N2, 0.5× ITSA, 1× GlutaMAX, 0.5× penicillin/streptomycin, and 50 µM 2-mercaptoethanol (Life Technologies). On DIV 2, media was additionally supplemented with a hedgehog inhibitor (1µM Cyclopamine) to inhibit ventralisation of the progenitors. On D7, Wnt inhibitor XAV-939 is removed. On DIV 11, progenitors were passaged and reseeded at 1:2.5. The following day, cultures were transferred to BDM containing 20 ng/mL fibroblast growth factor 2 (FGF2; R&D Systems) and 1µM Cyclopamine for 8 days. On DIV 19, progenitors were passaged and counted using the haemocytometer. Progenitors were cryopreserved or further differentiated into hippocampal progenitors and neurons.

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Differentiation into Hippocampal progenitors and neurons

Starting from DIV 19, cells were seeded at $0.45 \times 10^6/\text{cm}^2$ in basal differentiation medium with 50 nM chroman 1. After 24 hours, chroman 1 is removed and cells are maintained in basal differentiation media for 2 days. Media was then supplemented with 3 μM CHIR99021 (Wnt activator) for 10-14 days to generate hippocampal progenitors capable of differentiating into the different subregions of the hippocampus. Cells were then cultured in basal differentiation media only till DIV 40-45. Hippocampal progenitors are passaged 1:2 every 10-14 days into newly coated hLam521 plates.

For final plating, the plates or coverslips were coated with poly-L-ornithine followed by hLam521. Cells were then seeded at $2-3 \times 10^5/\text{cm}^2$ in cortex medium with 50 nM chroman 1. Media was then switched to maturation medium containing 1:1 DMEM/F12 and Neurobasal, $1 \times$ B27, $0.5 \times$ N2, $0.5 \times$ ITSA, $0.5 \times$ NEAA, $0.5 \times$ GlutaMAX, and $0.5 \times$ penicillin/streptomycin, with 10 ng/mL brain-derived neurotrophic factor, BDNF (R&D Systems), 10 μM γ -secretase inhibitor DAPT, 0.25 μM $N^6,2'$ -O-dibutyryl adenosine 3',5'-cyclic monophosphate (Tocris Bioscience), 200 nM ascorbic acid (Sigma-Aldrich). Cells were cultured for 3-7 days and DAPT was removed. Cells were further cultured until they were fixed with PFA for downstream experiment.

Generation and maturation of Hippocampal organoids

After the removal of CHIR99021 around DIV 33-35, cells were seeded at 50-60k per well of a 96 ultra-low attachment multiple well plate (Sigma- Aldrich- CLS7007) in CTX media with 25 nM chroman 1. After 48 hours, chroman 1 is removed and cells are maintained in CTX media for 14 days with media change every other 2 days. The organoids are transferred into a 10 cm Petri dish with maturation medium supplemented with 10 ng/mL brain-derived neurotrophic factor, BDNF (R&D Systems), 0.25 μM $N^6,2'$ -O-dibutyryl adenosine 3',5'-cyclic monophosphate (Tocris Bioscience), 200 nM ascorbic acid (Sigma-Aldrich), and cultured on an orbital shaker (Biotek-#NBT101SRC) in the incubator at 68 rpm.

Differentiation into hippocampal regionally specified astrocytes

Cells were harvested on DIV 19 using accutase and seeded at 5000 cells per well of a 96 ultra-low attachment multiple well plate (Sigma- Aldrich- CLS7007) in CTX media with FGF2. The plates are centrifuged at 1100 rpm for 5 minutes. Under these conditions, cells formed spheres after 24 hours and cultured for another 24 hours. Media was then changed to CTX media supplemented with 3 μM CHIR99021 for 14 days with media change every other 2 days. The spheres were then transferred into a 10cm petri dish and were maintained on an orbital shaker in the incubator. Spheres were cultured in CTX media only for 2 days before switching the media to astrodifferentiation medium (ADM) for a month and manually dissociated by cutting the spheres into 4 parts. ADM was comprised of Advanced DMEM/F12 (ThermoFisher Scientific, #12634-010) supplemented with 0.2% B-27 Supplement (ThermoFisher Scientific, #17504001), 0.5% N-2 Supplement (ThermoFisher

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Scientific, #17502001), 0.5% GlutaMAX, 1% ANTI-ANTI (ThermoFisher Scientific, #15240-062), as well as 20 ng/ml Human LIF Protein, HEK293 (Medchem Express, #HY-P73276) and 20 ng/ml EGF (ThermoFisher Scientific, #AF-100-15). After a month of culture in human LIF, LIF was replaced with human recombinant bFGF (StemCellTech, #78003) and cultured for additional 2-3 months. Spheres were manually dissociated by cutting the spheres into 4 parts every 2-3 weeks and transferred to a new petri dish.

Astrocyte expansion was achieved by dissociating the spheres with NeuroCult™ Enzymatic Dissociation Kit for Adult CNS Tissue (Mouse and Rat) (StemCellTech, #05715) and plated on Matrigel (Corning)-coated flask at 10^6 per T25 flask. The cells were maintained in the ADM supplemented with 20 ng/ml human recombinant bFGF and 20 ng/ml EGF (ThermoFisher Scientific, #AF-100-15). The cells were harvested at 70-80% confluency and reseeded at 0.5×10^5 per T75 flask and maintained in ADM supplemented with 20 ng/ml human recombinant bFGF and 20 ng/ml EGF. To confirm the absence of neuronal contamination, cells were seeded into 96-well plates and subjected to immunostaining for MAP2 and GFAP. To ensure biological reproducibility, independent replicates were generated by maintaining separate neurospheres for at least one month prior to initiating terminal differentiation protocols.

Immunocytochemistry, microscopy, and quantification

Immunocytochemistry was performed as previously described (76–78) (see Supplementary Material 1 for a list of primary and secondary antibodies). Briefly, cells were fixed for 10 minutes in 4% paraformaldehyde. Cells were permeabilised and blocked in 2% donkey serum, 0.1% Triton-X in PBS-/- with sodium azide (Blocking and permeabilisation buffer) for 20-30 minutes at room temperature. Fixed cells with primary antibodies were incubated at 4°C in the blocking and permeabilisation buffer overnight. Secondary antibodies were incubated on the fixed cells for 1-2 hours at room temperature. After PBS washes and nuclear counterstaining with Hoechst 33342, cells were mounted in Fluoromount-G™ Mounting Medium (#00-4958-02). Mounted cells were stored at 4°C before imaging. Images were captured on either an EVOS epifluorescence microscope, Operetta High Content Imaging or a Zeiss LSM900 confocal microscope. For quantification of cell numbers and markers, images across wells were generated with at least five fields of view from multiple wells. Images were counted using the CellProfiler software. All data visualizations were generated using custom Python scripts, and statistical analyses were performed using the SciPy stats library (Python v3.10).

For synapse quantification, images were processed using the SynBot macro as previously described (66). Images were converted to RGB and using the SynBot macro with the default parameters of the manual thresholding method, we quantified synapse numbers.

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Gene expression analysis

For qPCR, cells were harvested at different stages of differentiation and centrifuged at 200 *g* for 5 minutes. After centrifugation, the supernatant was removed, and the cell pellet was resuspended in 100 mL of RNeasy lysis buffer (ThermoFisher Scientific # AM7020) and stored at -80°C. RNA was extracted using an RNeasy Mini Kit (ThermoFisher Scientific # 12183018A). After extraction, RNA was quantified using the NanoDrop 2000 UV Visible Spectrophotometer. RNA samples were then stored at -80°C or used for downstream qPCR analysis using the PrimeTime™ One-Step 4X Broad-Range Master Mix (Integrated DNA Technologies #10011746). We used 8-10 ng of RNA for each reaction using in-house designed primers and probes under the following conditions: 55°C for 10 minutes, 95°C for 1 minute, and 40 cycles of 95°C for 15 seconds and 60°C for 60 seconds. Ct values were normalised to housekeeping gene GAPDH. All gene expression data were determined using $2^{-\Delta Ct}$ from at least three independent biological replicates. Table S1 presents a list of all primers.

Multi-electrode assay (MEA)

For electrophysiological studies, monolayer hippocampal neurons and organoids (from CLV2-10 and CLV3-1) were transferred onto poly-D-Lysine (PDL)/laminin-521-coated MEA chips (Multi Channel System). MEA chips were coated with PDL and human Laminin-521 at a concentration of 50 µg/ml and 10 µg/ml respectively. Monolayer hippocampal neurons were dissociated at day 40 and plated on PDL/laminin-521-coated MEA chips at a density of 5×10^5 cells per MEA chip. To assess the functional properties of human pluripotent stem cell (hPSC)-derived astrocytes, we co-cultured hippocampal neurons (DIV 40) with passage 8 (P+8) hippocampal astrocytes on MEA chips. Based on published data (79,80), we cocultured the neurons and astrocytes at a ratio of 1.4:1 (neurons:astrocytes). The organoids at DIV 55 were transferred onto PDL/laminin-521-coated chips.

The MEA chips used for this study have 64 electrodes. All recordings were conducted using a CL1 system (v.0.8, Cortical Labs, Singapore) maintained at 37°C (81). The recording of the baseline activity of MEA chips was performed weekly over a 5-minute period for 15-20 weeks after plating.

MEA analysis

The following electrophysiological parameters were analyzed: mean firing rate in Hz, max firing rate in Hz, variance of the firing in Hz, criticality analysis, power law observables, branching ratio, exponent relationship, Deviation from Criticality Coefficient (DCC), information entropy, and burst detection analysis which assessed both the number of bursts, the average number of spikes during bursts, and the percentage of bursting activity overall. The analysis for criticality analysis, power law observables, branching ratio, exponent relation, and DCC were partially adapted from previously published studies (72).

Criticality analysis

Avalanche analysis was used to characterise how close the network operates to a

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critical state. Avalanche onsets and terminations were defined based on crossings of a predefined threshold in the global network activity signal (79). Individual avalanches may be triggered by spiking activity from any subset of neurons within the region of interest. For each avalanche, both the total number of spikes involved (S) and the overall event duration (D) were quantified. To assess the proximity of the cultured neuronal networks to criticality, we examined the presence of several established dynamical signatures in the data: (1) power-law behaviour of avalanche observables; (2) consistency of scaling exponent relationships; (3) the value of the branching ratio; and (4) the existence of an appropriate scaling function. These indicators constitute necessary conditions for critical dynamics and jointly satisfying them provides strong evidence that the system operates near a critical point (80,81).

Power law observables

A system operating at criticality consists of interacting elements, in this case the recorded electrodes, that exhibit ongoing fluctuations in activity while retaining a degree of coordination across individual units, such as correlated spiking. Critical dynamics are characterised by scale-invariant behaviour, whereby events across both temporal and spatial scales follow power-law statistics (82–84). In the networks studied here, such events correspond to continuous cascades of spiking activity, as opposed to isolated local bursts or globally synchronous network-wide discharges. These cascades are commonly referred to as *neuronal avalanches*. To examine this phenomenon in our recordings, we analysed binarised spike trains from each electrode. Each recording session was segmented into temporal bins of 50 ms, and network activity at each timestep was defined as the sum of spiking activity across all neurons within that bin. An activity threshold was then set at 40% of the median network activity computed across all time bins. Avalanche initiation and termination were identified as the moments when the network activity crossed this threshold from below and subsequently returned below it (84). Importantly, the results were stable across a broad range of threshold values spanning 30% to 70% of the median activity. The size of an avalanche, S , is the total number of spikes during the avalanche. The avalanche duration, D , is the time between threshold crossings. Following the approach in (79), a truncated power-law model was fitted to the avalanche size distribution using maximum likelihood estimation as below:

$$f(S) = \frac{S^{-\tau}}{\sum_{S_{min}}^{S_{max}} S^{-\tau}} \quad (1)$$

Here, τ denotes the power law exponent associated with the avalanche size distribution. For a given neuronal recording session in which N_A avalanches are identified, the estimation of this model follows an iterative fitting process as described in (85), as outlined below:

1. Find the maximum observed avalanche size S_{max} .
2. Evaluate the three different power law exponents, τ , for the 3 smallest avalanche sizes observed, S_{min} .

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3. Calculate the *Kolmogorov-Smirnov* (KS) to assess the goodness-of-fit between the fitted power-law model and the empirical distribution.
4. Select the minimum KS value among all estimates, along with the corresponding τ and S_{min} values.
5. Complete the estimation if $KS < \frac{1}{\sqrt{N_A}}$ otherwise repeat steps 2 to 5 with S_{max} reduced by 1 until this condition is met.

Steps 3–5 ensure that the observed data distribution is consistent with a power-law model rather than alternative heavy-tailed candidates, such as log-normal or stretched-exponential distributions (86).

The same analysis was then carried out for the set of avalanche durations DDD, yielding the corresponding power-law exponent α for the distribution of avalanche durations.

Exponent relation and Deviation from Criticality Coefficient (DCC)

In systems operating near criticality, an additional scaling relationship links the power-law exponents α and τ to the exponent governing the mean avalanche size ($\langle S \rangle$), as a function of avalanche duration D (87). We first estimated the third scaling exponent of the system, β , directly from the experimental data using linear regression, assuming the following exponent relationship holds for a system at criticality:

$$\langle S \rangle \propto D^{\beta_{fit}} \quad (2)$$

This third scaling exponent further connects the avalanche size and duration distributions and is theoretically given by:

$$\beta_{pred} = \frac{(\alpha - 1)}{(\tau - 1)} \quad (3)$$

Comparing the fitted value from the empirical data (β_{fit}) and its estimation using α and τ exponents, (β_{pred}), a new measure is derived to evaluate the *Deviation from Criticality Coefficient* (DCC), parameterized as dcc:

$$dcc = \| \beta_{pred} - \beta_{fit} \| \quad (4)$$

where, β_{pred} and β_{fit} are the predicted and fitted values of, β respectively.

Accordingly, lower DCC values correspond to a closer agreement between the fitted power-law model and the empirical distribution.

Information Entropy:

A binary vector was used to represent the presence or absence of a spike in each 50 ms time bin for each recorded electrode. We then calculated the binary entropy function for every electrode using their binary spike trains.

The binary entropy function, denoted $H(p)$ or $H_b(p)$, is defined as the entropy of a Bernoulli process with probability p of one of two outcomes.

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Given that:

- $Pr(X = 1) = p$
- $Pr(X = 0) = 1 - p$

The entropy of X (measured in shannons) is calculated as:

$$H(X) = H_b(p) = -p \cdot \log_2(p) - (1 - p) \cdot \log_2(1 - p)$$

where $0 \cdot \log_2(0)$ is defined to be 0.

Hence, we calculated the local entropy each electrode in each time bin and averaged across time and electrodes to compare the groups of in vitro cultures.

Burst detection analysis

To identify bursting activity, we analysed spike trains from each electrode. A burst was defined as a sequence of at least N spikes (default: 5) in which the inter-spike interval (ISI) between consecutive spikes did not exceed a fixed threshold (default: 1 ms). For each electrode, spike times were first sorted, and bursts were detected by scanning for sequences of spikes that satisfied both the minimum spike count and maximum ISI criteria. For each detected burst, we recorded its duration (time between the first and last spike, in seconds) and size (total number of spikes). These burst metrics were stored for downstream analysis.

Construction of Causal Connectivity Graphs Using Granger Causality

To infer directed causal relationships among activities of recorded electrodes, we constructed causal connectivity graphs from spike train recordings using Granger causality analysis (86). Recordings were obtained from 64-channel multi-electrode arrays (MEAs) sampled at 25 kHz. After excluding the reference electrodes (electrode IDs 0, 4, 7, 56, and 63), a total of 59 active electrodes were retained for analysis.

Spike times were first extracted from each recording and converted into binary spike matrices ("spikewords") representing binned firing activity for each channel. The bin length for this analysis was set to 1 s. Each matrix entry indicated whether a spike occurred in a given bin for a given electrode. Electrodes with mean firing rates below 0.05 Hz were excluded to reduce noise from inactive channels.

Pairwise Granger causality tests were then computed between all remaining electrodes to estimate directional dependencies in spiking activity. For each ordered electrode pair (i, j) , the corresponding time series were concatenated into a two-column matrix $[x_i, x_j]$, and Granger causality tests were performed across lags from 1 to 50 using the F-test formulation. The minimum p -value across lags was used to determine significance ($p < 0.05$), and the corresponding F-statistic was recorded as

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the causal connection strength. Connections failing to meet this significance criterion were omitted.

The resulting weighted, directed adjacency matrix $A \in \mathbb{R}^{\{64 \times 64\}}$ was constructed such that $A_{\{ij\}}$ represents the causal influence from electrode j to electrode i . Self-connections were excluded by enforcing $A_{\{ii\}} = 0$. The number of incoming and outgoing connections was also computed for each electrode to characterize its causal in-degree and out-degree.

To quantify the topological and dynamical organization of these causal networks, several standard graph-theoretical measures were computed from each adjacency matrix.

- **The total number of edges (E)** was calculated as the number of significant directed causal links detected in the network.

The average edge weight ($\bar{w}$) was obtained as the mean F-statistic across all significant connections:

$$\bar{w} = \frac{1}{E} \sum_{(i,j) \in E} w_{ij}$$

- **The clustering coefficient** was computed using the weighted variant of the Watts–Strogatz formulation (87) as extended by Onnela et al. (88). For an undirected and weighted graph $G = (V, E)$ with edge weights $w_{ij} \geq 0$, the local weighted clustering coefficient C_i^w for a node i is defined as:

$$C_i^w = \frac{1}{k_i(k_i - 1)} \sum_{j,k} (\hat{w}_{ij} \hat{w}_{ik} \hat{w}_{kj})^{\frac{1}{3}}$$

where k_i is the degree of node i (number of neighbors), $\hat{w}_{ij} = w_{ij} / \max(w)$ is the normalized edge weight, and $(\hat{w}_{ij} \hat{w}_{ik} \hat{w}_{kj})^{\frac{1}{3}}$ represents the geometric mean of the weights forming a triangle between nodes i, j , and k . The global clustering coefficient for the entire network is then computed as:

$$C^w = \frac{1}{N} \sum_{i \in V} C_i^w$$

This metric ranges from 0 (no clustering) to 1 (perfect clustering), providing a normalized measure of how locally interconnected the network is. A higher C^w indicates that groups of electrodes tend to form tightly connected subnetworks, suggesting strong local synchronization or recurrent information flow among nearby neurons, whereas lower values imply more random or sparse local interactions.

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- **The mean betweenness centrality** was calculated as the average of the weighted betweenness scores across all nodes
Mathematically, the betweenness centrality $C_B(v)$ of a node v is defined as:

$$C_B(v) = \sum_{\substack{s, t \in V \\ s \neq v \neq t}} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the total number of shortest paths between nodes s and t , and $\sigma_{st}(v)$ is the number of those paths that pass through node v . For weighted networks, edge lengths are defined as the inverse of their weights ($\ell_{ij} = 1/w_{ij}$), such that stronger causal connections corresponded to shorter path distances. The mean betweenness centrality across all $N = |V|$ nodes was computed as:

$$\overline{C_B} = \frac{1}{N} \sum_{v \in V} C_B(v)$$

This measure captures how strongly specific electrodes act as communication hubs that mediate information flow between otherwise disconnected neuronal populations. A high mean betweenness centrality therefore indicates a more centralized network organization dominated by a few influential nodes that integrate and redistribute signals, whereas lower values suggest a more distributed and parallel mode of information transfer across the culture.

Statistical Analysis (Group Comparisons)

To examine for statistical differences between cell lines in immunocytochemistry and quantitative PCR measurements, a two-tailed Mann-Whitney U test was performed, and group values were presented as mean $\pm$ standard error of the mean (SEM). To compare organoids and 2D cultures across multiple activity metrics in electrophysiological and causal connectivity analysis, outliers were removed based on the interquartile range (IQR) of culture-level mean firing rates within each group. For each metric, group means and standard errors were visualized using box plots. Statistical differences between groups were assessed using Welch's ANOVA, followed by a Games–Howell post-hoc test to account for unequal variances and sample sizes. Significance thresholds were annotated as follows: *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.

Survival Analysis

To compare the time-to-event distributions between electrophysiological and functional connectivity metrics, the log-rank (Mantel–Cox) test was used to assess statistical differences in survival curves, which were visualized using Kaplan–Meier plots. Statistical significance was defined as $p < 0.05$.

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Temporal Statistical Analysis

Temporal differences in immunocytochemistry and quantitative PCR measurements within cell lines were assessed using a Wilcoxon signed-rank test, followed by Bonferroni correction to account for multiple comparisons. Differences between cell lines at each time point were assessed using the Mann–Whitney U test. Significance thresholds were annotated as follows: *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.

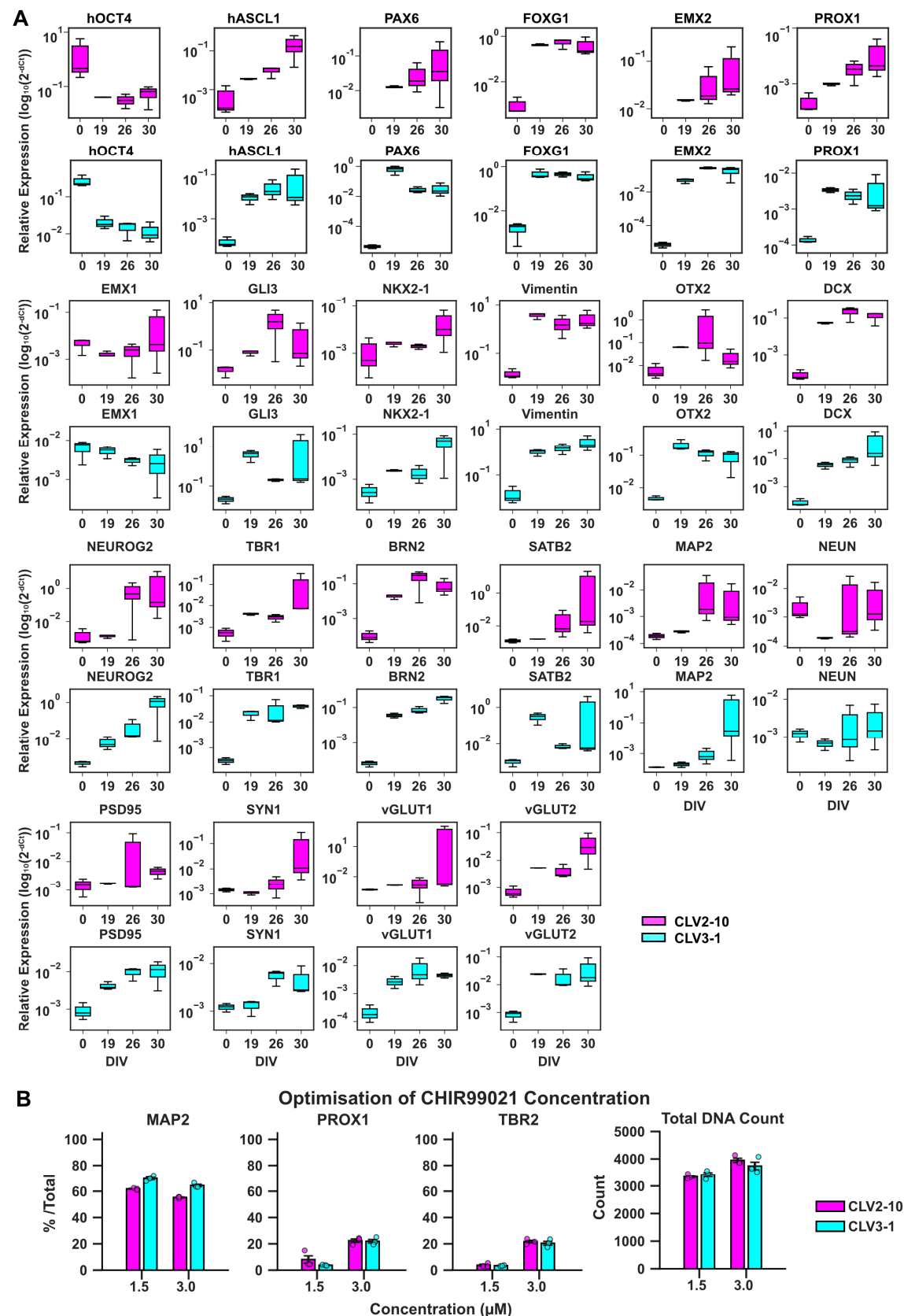
Supplementary Data and Methods

Supplementary data consists of **Supplementary Figure 1 – 5** included below for convenience with high resolution images available to download.

Supplementary Tables 1 – 29 with full results from all statistical analysis are available to download as a separate file.

Supplementary Methods are available to download as a separate file.

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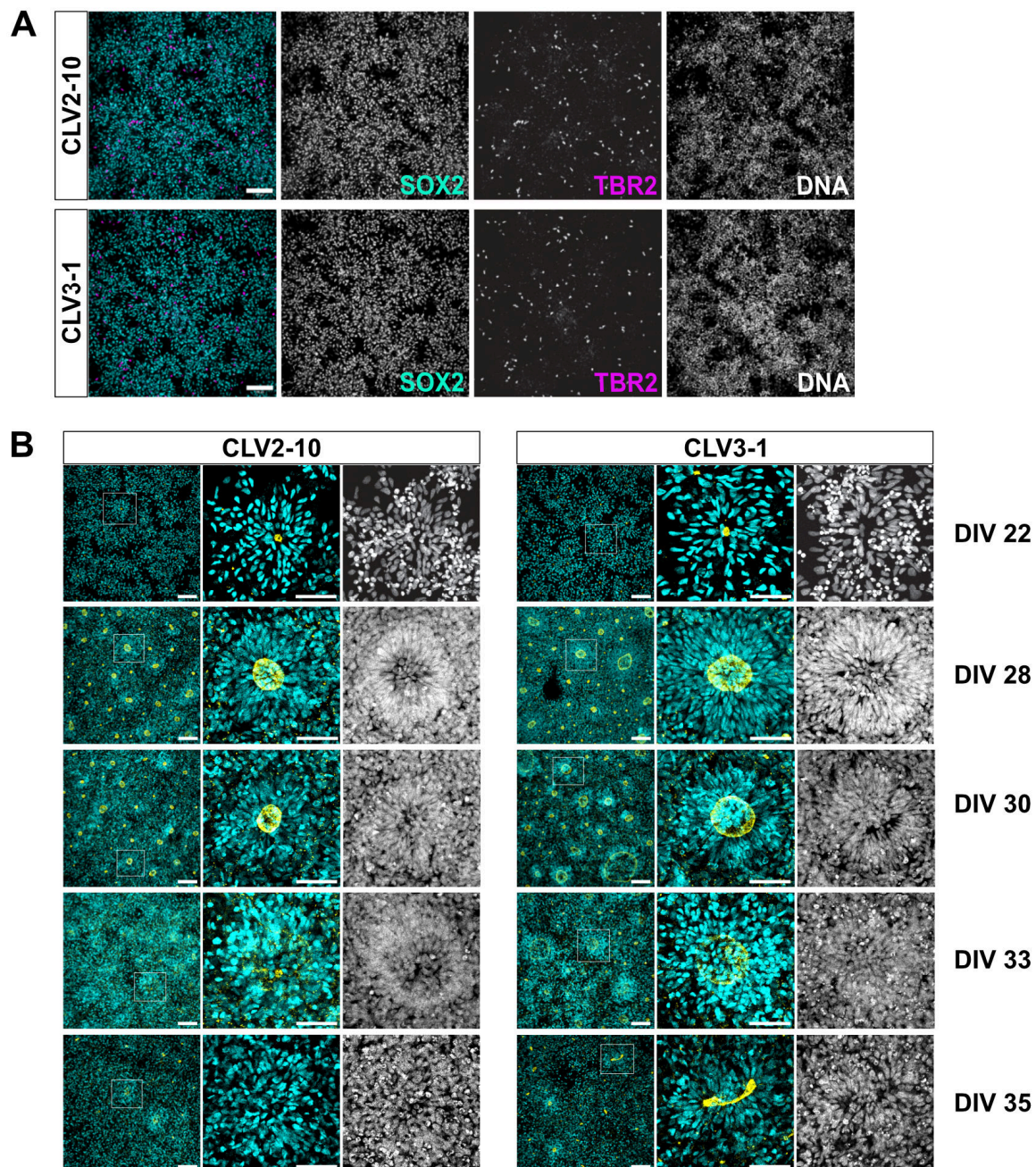


Supplementary Figure 1:

A- qRT-PCR analysis confirming the increased expression of neuronal markers and reduced expression of pluripotency markers for two hPSC lines (male and female, $n = 3$ independent experiments each). Boxplots with no data points indicate that no measurements were obtained for that marker on the given day in vitro.

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B- Quantification for PROX1+ and TBR2+ cell populations to validate the optimised CHIR99021 concentration at DIV40 for two hPSC lines (male and female, $n = 3$ independent experiments each). The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

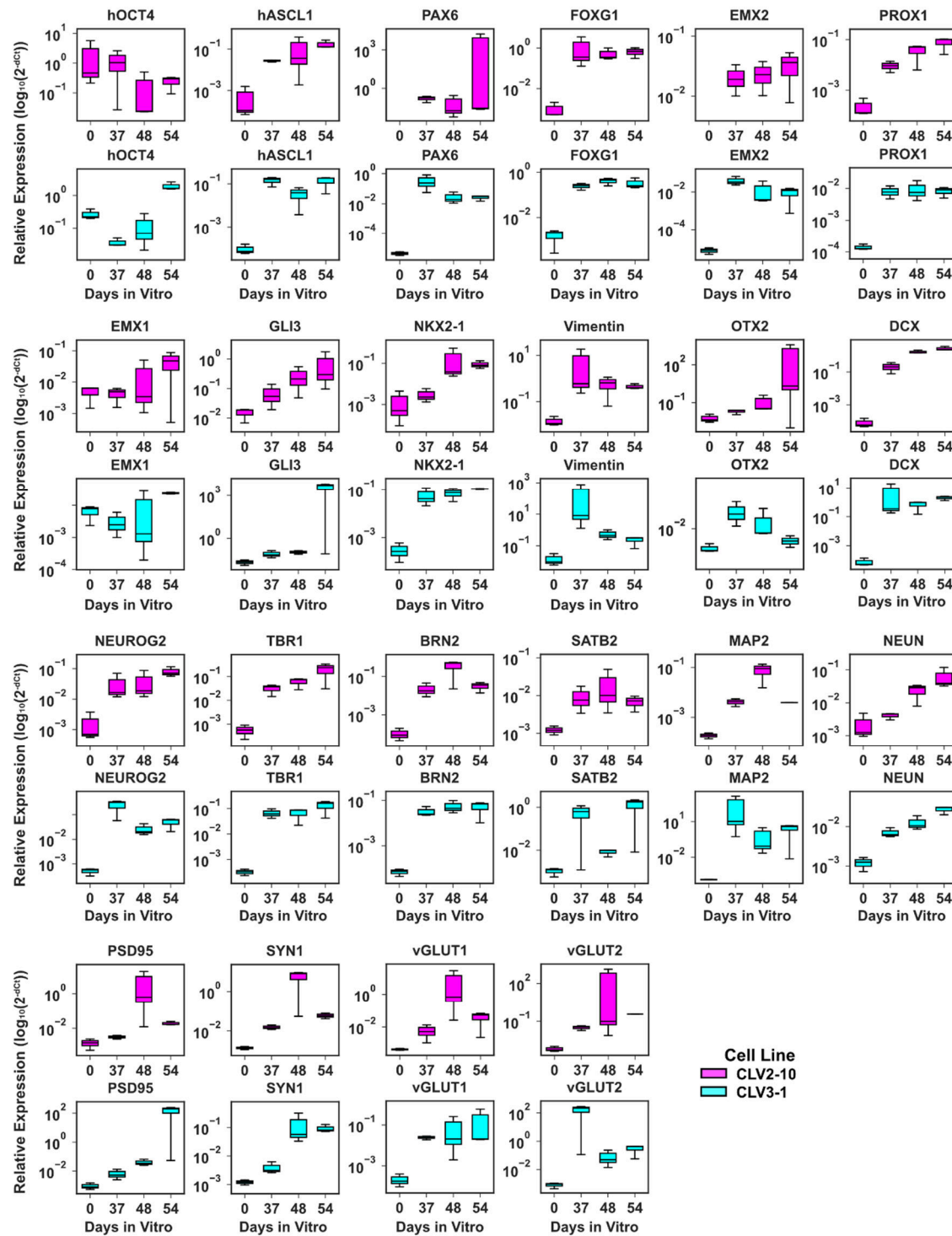


Supplementary Figure 2:

A- Immunostaining analysis of SOX2+ progenitors and TBR2+ IPCs at DIV 22 for two hPSC lines (male and female, $n = 3$ independent experiments each). Scale bar is 100 μm .

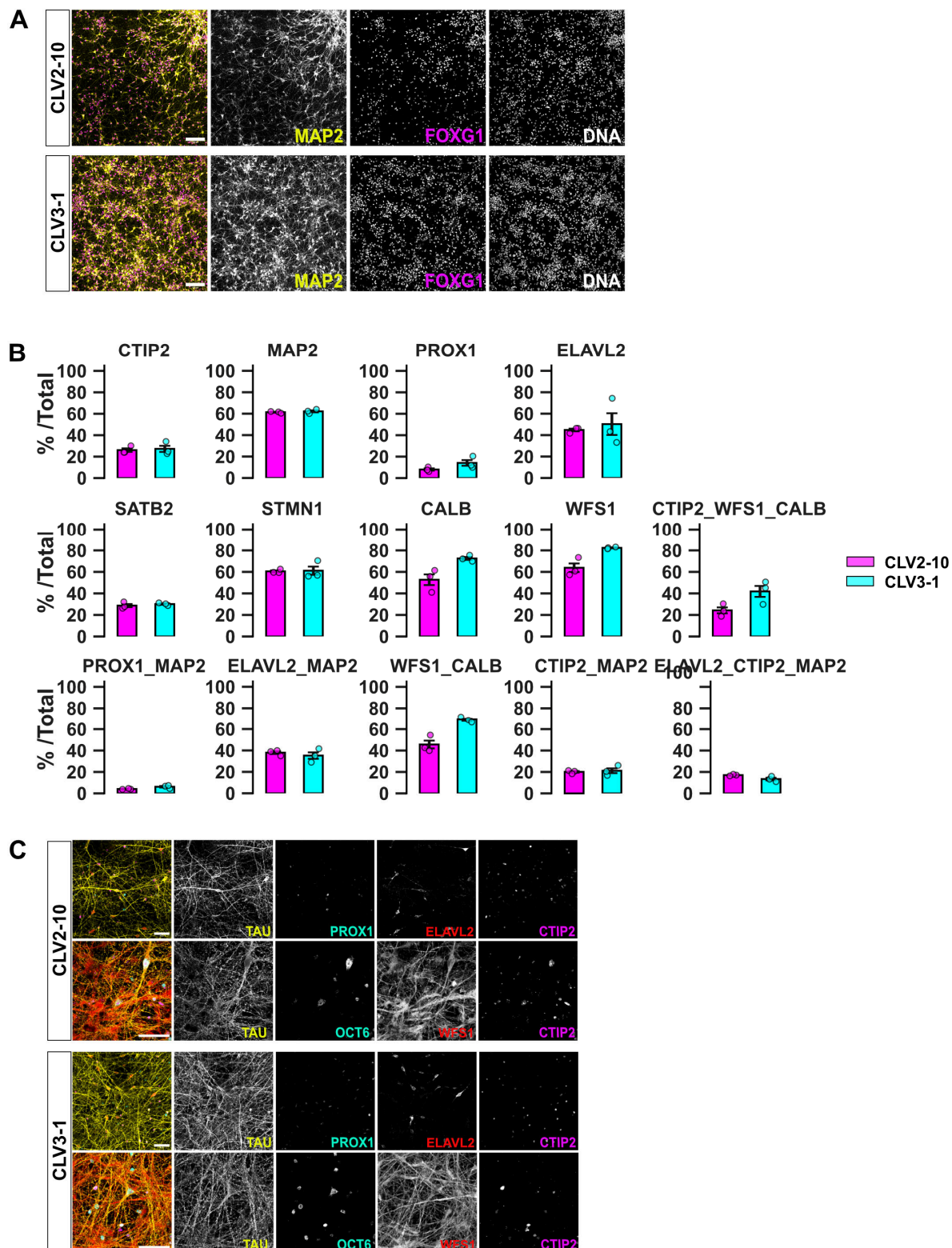
B- Immunostaining analysis of SOX2+ progenitors and ZO-1+ lumens showing the gradual increase in diameter of the lumen for two hPSC lines (male and female, $n = 3$ independent experiments each). Scale bar is 100 μm and 50 μm .

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Supplementary Figure 3: qRT-PCR analysis confirming the increased expression of neuronal markers and reduced expression of pluripotency markers for two hPSC lines (male and female, $n = 3$ independent experiments each). Boxplots with no data points indicate that no measurements were obtained for that marker on the given day in vitro.

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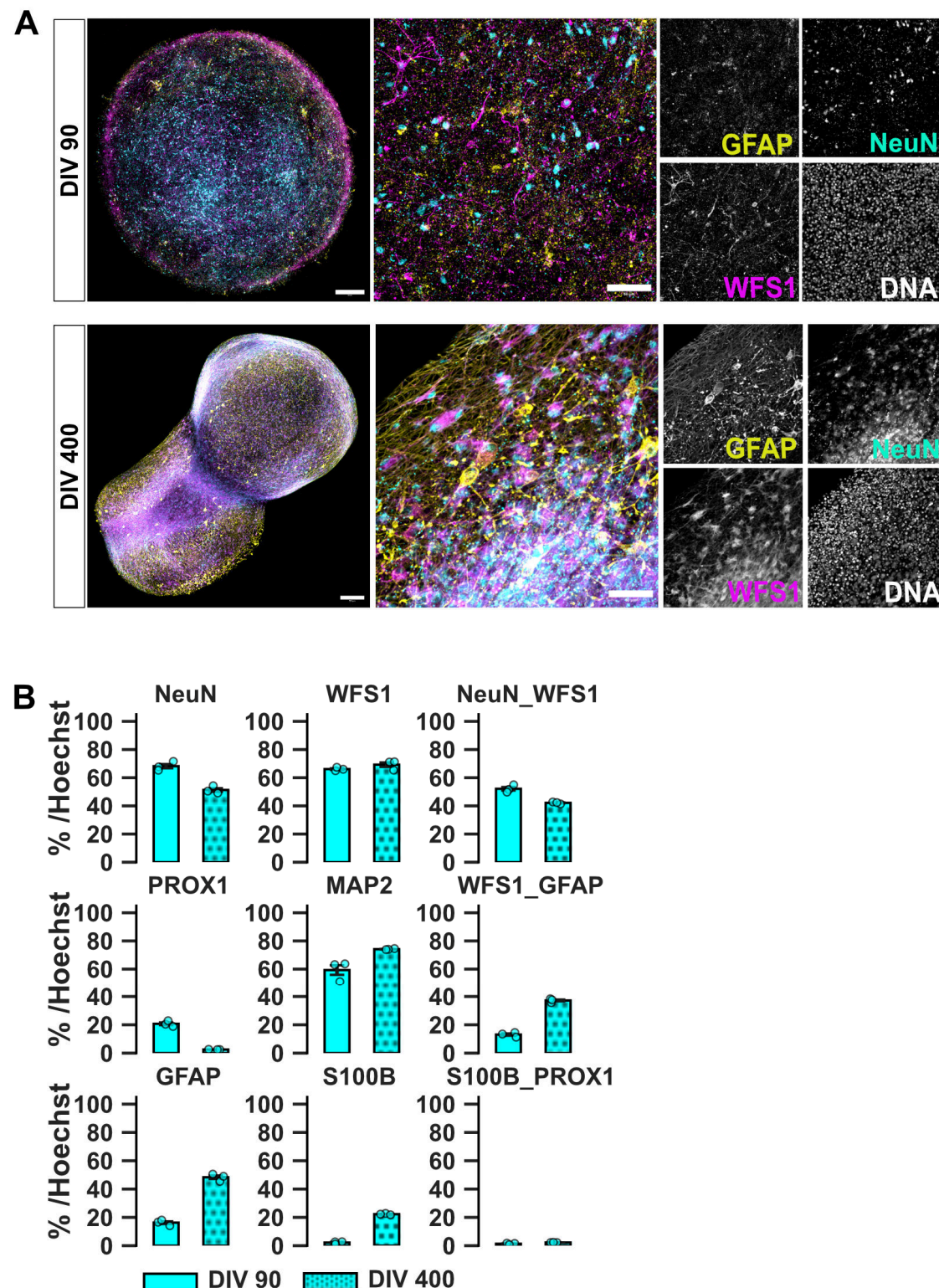
Supplementary Figure 4:

A- Representative images of FOXG1+ and MAP2+ neurons at DIV 60. Scale bar is 100 μ m.

B- Quantification of CTIP2+, MAP2+, PROX1+, ELAVL2+, SATB2+, STMN1+, CALB+ and WFS1+ cell populations at DIV 60 for two hPSC lines (male and female, n = 3 independent experiments each, data are mean $\pm$ SEM). The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

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C- Immunostaining analysis of TAU+, ELAVL2+, PROX1+, OCT6, WFS1+ and CTIP2+ neurons showing large somata pyramidal neurons and small somata neurons at DIV 90. Scale bar is 50 μ m.



Supplementary Figure 5:

A- Immunostaining analysis of DIV 90 and DIV 400 HOs showing the presence of astrocytes, GFAP+ cells and NeuN+ and WFS1+ neurons. Note that the organoids at DIV400 are two separate organoids that fused during culture.

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B- Quantification of astrocyte-neuronal composition of HOs at DIV 90 and DIV 400 ($n = 3$ independent experiments, data are mean $\pm$ SEM). The Y-axis refers to the % of cells expressing the marker over the total of cells counted as per Hoechst staining.

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Author contributions

K.D.A.B and B.J.K conceived and designed the experiments. K.D.A.B and C.D performed neuronal differentiation and other cellular and molecular assays. Data analysis and interpretation were carried out by K.D.A.B., C.D. and H.W.C. Electrophysiological recordings and data acquisition were conducted by K.D.A.B. and C.D., while electrophysiological data analysis was performed by F.H. and H.W.C. The manuscript was written and revised by K.D.A.B., C.D., F.H., H.W.C., B.W., M.D., and B.J.K. The work was supervised by B.J.K. All authors discussed, commented and agreed on the manuscript.

Competing interests

The authors declare the following competing interests: K.D.A.B, C.D., F.H., H.W.C., and B.K. are employed by Cortical Labs Pty Ltd, a for-profit company interested in the commercial viability of synthetic biological intelligence and related patents. K.D.A.B, C.D., F.H., H.W.C., and B.K. also hold a pecuniary interest in Cortical Labs Pty Ltd.

RESEARCH ARTICLE

References

1. Bird CM, Burgess N. The hippocampus and memory: insights from spatial processing. *Nat Rev Neurosci*. 2008 Mar;9(3):182–94.
2. Eichenbaum H. Hippocampus: Cognitive Processes and Neural Representations that Underlie Declarative Memory. *Neuron*. 2004 Sept 30;44(1):109–20.
3. Lisman J, Buzsáki G, Eichenbaum H, Nadel L, Ranganath C, Redish AD. Viewpoints: how the hippocampus contributes to memory, navigation and cognition. *Nat Neurosci*. 2017 Nov;20(11):1434–47.
4. Nakazawa K, Quirk MC, Chitwood RA, Watanabe M, Yeckel MF, Sun LD, et al. Requirement for Hippocampal CA3 NMDA Receptors in Associative Memory Recall. *Science*. 2002 July 12;297(5579):211–8.
5. Paulson AL, Zhang L, Prichard AM, Singer AC. 40 Hz sensory stimulation enhances CA3-CA1 coordination and prospective coding during navigation in a mouse model of Alzheimer's disease [Internet]. *Neuroscience*; 2024 [cited 2026 Jan 4]. Available from: <http://biorxiv.org/lookup/doi/10.1101/2024.10.23.619408>
6. Gao Y, Qiao Y, Wang X, Zhu M, Yu L, Yuan H, et al. Downregulation of Neuralized1 in the Hippocampal CA1 Through Reducing CPEB3 Ubiquitination Mediates Synaptic Plasticity Impairment and Cognitive Deficits in Neuropathic Pain. *Neurosci Bull*. 2025 Dec;41(12):2233–53.
7. Toda T, Parylak S, Linker SB, Gage FH. The role of adult hippocampal neurogenesis in brain health and disease. *Mol Psychiatry*. 2019 Jan;24(1):67–87.
8. Yoshimura S, Takagi Y, Harada J, Teramoto T, Thomas SS, Waeber C, et al. FGF-2 regulation of neurogenesis in adult hippocampus after brain injury. *Proc Natl Acad Sci*. 2001 May 8;98(10):5874–9.
9. Sakaguchi H, Kadoshima T, Soen M, Narii N, Ishida Y, Ohgushi M, et al. Generation of functional hippocampal neurons from self-organizing human embryonic stem cell-derived dorsomedial telencephalic tissue. *Nat Commun*. 2015 Nov 17;6(1):8896.
10. Khalaf-Nazzal R, R. Khalaf-Nazzal, Francis F, F. Francis. Hippocampal development – Old and new findings. *Neuroscience*. 2013 Sept 17;248(248):225–42.
11. Roy DS, Park YG, Kim ME, Zhang Y, Ogawa SK, DiNapoli N, et al. Brain-wide mapping reveals that engrams for a single memory are distributed across multiple brain regions. *Nat Commun*. 2022 Apr 4;13(1):1799.
12. Jimenez JC, Berry JE, Lim SC, Ong SK, Kheirbek MA, Hen R. Contextual fear memory retrieval by correlated ensembles of ventral CA1 neurons. *Nat Commun*. 2020 July 13;11(1):3492.
13. Atucha E, Ku SP, Lippert MT, Sauvage MM. Recalling gist memory depends on CA1 hippocampal neurons for lifetime retention and CA3 neurons for memory precision. *Cell Rep*. 2023 Nov;42(11):113317.
14. Witter MP. Hippocampal Formation. In: *Encyclopedia of Neuroscience* [Internet]. Springer, Berlin, Heidelberg; 2009 [cited 2025 Dec 22]. p. 1840–5. Available from: https://link.springer.com/rwe/10.1007/978-3-540-29678-2_2212
15. Leonardis EJ. Hippocampus. In: *Encyclopedia of Animal Cognition and Behavior* [Internet]. Springer, Cham; 2022 [cited 2025 Dec 22]. p. 3117–23. Available from: https://link.springer.com/rwe/10.1007/978-3-319-55065-7_1267
16. Akyüz E, Aslan FŞ. Role of the Hippocampus in Cognition. In: Turgut M, Acer N, Turgut AT, Di Rocco C, editors. *The Human Hippocampus: Development, Neuroanatomy, Neurophysiology, Neuropathology and Surgery* [Internet]. Cham: Springer Nature Switzerland; 2025 [cited 2025 Dec 22]. p. 205–14. Available from: https://doi.org/10.1007/978-3-031-88907-3_15
17. Beyaz F. Structural and Functional Properties of Hippocampal Neurons. In: Turgut M, Acer N, Turgut AT, Di Rocco C, editors. *The Human Hippocampus: Development, Neuroanatomy, Neurophysiology, Neuropathology and Surgery* [Internet]. Cham: Springer Nature Switzerland; 2025 [cited 2025 Dec 22]. p. 23–37. Available from: https://doi.org/10.1007/978-3-031-88907-3_4
18. Takahashi K, Tanabe K, Ohnuki M, Narita M, Ichisaka T, Tomoda K, et al. Induction of Pluripotent Stem Cells from Adult Human Fibroblasts by Defined Factors. *Cell*. 2007 Nov 30;131(5):861–72.
19. Sakaguchi H, Kadoshima T, Soen M, Narii N, Ishida Y, Ohgushi M, et al. Generation of functional hippocampal neurons from self-organizing human embryonic stem cell-derived dorsomedial telencephalic tissue. *Nat Commun*. 2015 Nov 17;6(1):8896.
20. Pomeshchik Y, Klementieva O, Gil J, Martinsson I, Hansen MG, De Vries T, et al. Human iPSC-Derived Hippocampal Spheroids: An Innovative Tool for Stratifying Alzheimer Disease Patient-Specific Cellular Phenotypes and Developing Therapies. *Stem Cell Rep*. 2020 July;15(1):256–73.

RESEARCH ARTICLE

21. Sarkar A, Mei A, Paquola ACM, Stern S, Bardy C, Klug JR, et al. Efficient Generation of CA3 Neurons from Human Pluripotent Stem Cells Enables Modeling of Hippocampal Connectivity In Vitro. *Cell Stem Cell*. 2018 May 3;22(5):684-697.e9.
22. Yu DX, Di Giorgio FP, Yao J, Marchetto MC, Brennand K, Wright R, et al. Modeling Hippocampal Neurogenesis Using Human Pluripotent Stem Cells. *Stem Cell Rep*. 2014 Mar;2(3):295-310.
23. Wu Y, Cheng J, Qi J, Hang C, Dong R, Low BC, et al. Three-dimensional liquid metal-based neuro-interfaces for human hippocampal organoids. *Nat Commun*. 2024 May 14;15(1):4047.
24. Chambers SM, Fasano CA, Papapetrou EP, Tomishima M, Sadelain M, Studer L. Highly efficient neural conversion of human ES and iPS cells by dual inhibition of SMAD signaling. *Nat Biotechnol*. 2009 Mar;27(3):275-80.
25. Nishimura H, Li Y. Human pluripotent stem cell-derived models of the hippocampus. *Int J Biochem Cell Biol*. 2024 Dec;177:106695.
26. Alia C, Terrigno M, Busti I, Cremisi F, Caleo M. Pluripotent Stem Cells for Brain Repair: Protocols and Preclinical Applications in Cortical and Hippocampal Pathologies. *Front Neurosci*. 2019 Aug 6;13:684.
27. A dynamic gradient of Wnt signaling controls initiation of neurogenesis in the mammalian cortex and cellular specification in the hippocampus. *Dev Biol*. 2007 Nov 1;311(1):223-37.
28. Dunville K, Tonelli F, Novelli E, Codino A, Massa V, Frontino AM, et al. Laminin 511 and WNT signalling sustain prolonged expansion of hiPSC-derived hippocampal progenitors. *Development*. 2022 Sept 9;149(20):dev200353.
29. Grove EA, Tole S, Limon J, Yip L wah, Ragsdale CW. The hem of the embryonic cerebral cortex is defined by the expression of multiple Wnt genes and is compromised in Gli3-deficient mice. *Development*. 1998 June 15;125(12):2315-25.
30. Lee SMK, Tole S, Grove EA, McMahon AP. A local Wnt-3a signal is required for development of the mammalian hippocampus. *Development*. 2000 Feb 1;127(3):457-67.
31. Knight HM, Walker R, James R, Porteous DJ, Muir WJ, Blackwood DHR, et al. GRIK4/KA1 protein expression in human brain and correlation with bipolar disorder risk variant status. *Am J Med Genet B Neuropsychiatr Genet*. 2012 Jan 1;159B(1):21-9.
32. Sarkar A, Mei A, Paquola ACM, Stern S, Bardy C, Klug JR, et al. Efficient Generation of CA3 Neurons from Human Pluripotent Stem Cells Enables Modeling of Hippocampal Connectivity In Vitro. *Cell Stem Cell*. 2018 May;22(5):684-697.e9.
33. From Cradle to Grave: The Multiple Roles of Fibroblast Growth Factors in Neural Development. *Neuron*. 2011 Aug 25;71(4):574-88.
34. Ohkubo Y, Uchida AO, Shin D, Partanen J, Vaccarino FM. Fibroblast Growth Factor Receptor 1 Is Required for the Proliferation of Hippocampal Progenitor Cells and for Hippocampal Growth in Mouse. *J Neurosci*. 2004 July 7;24(27):6057-69.
35. Man JHK, Breur M, Van Gelder CAGH, Marcon G, Maderna E, Giaccone G, et al. Region-specific and age-related differences in astrocytes in the human brain. *Neurobiol Aging*. 2024 Aug;140:102-15.
36. Viana JF, Machado JL, Abreu DS, Veiga A, Barsanti S, Tavares G, et al. Astrocyte structural heterogeneity in the mouse hippocampus. *Glia*. 2023;71(7):1667-82.
37. Buosi AS, Matias I, Araujo APB, Batista C, Gomes FCA. Heterogeneity in Synaptogenic Profile of Astrocytes from Different Brain Regions. *Mol Neurobiol*. 2018 Jan;55(1):751-62.
38. Haim LB, Rowitch DH. Functional diversity of astrocytes in neural circuit regulation. *Nat Rev Neurosci*. 2017 Jan;18(1):31-41.
39. Zuchero JB, Barres BA. Glia in mammalian development and disease. *Development*. 2015 Nov 15;142(22):3805-9.
40. Karpf J, Unichenko P, Chalmers N, Beyer F, Wittmann MT, Schneider J, et al. Dentate gyrus astrocytes exhibit layer-specific molecular, morphological and physiological features. *Nat Neurosci*. 2022 Dec;25(12):1626-38.
41. Sultan S, Li L, Moss J, Petrelli F, Cassé F, Gebara E, et al. Synaptic Integration of Adult-Born Hippocampal Neurons Is Locally Controlled by Astrocytes. *Neuron*. 2015 Dec 2;88(5):957-72.
42. Guttikonda SR, Sikkema L, Tchieu J, Saurat N, Walsh RM, Harschnitz O, et al. Fully defined human pluripotent stem cell-derived microglia and tri-culture system model C3 production in Alzheimer's disease. *Nat Neurosci*. 2021 Mar;24(3):343-54.
43. Gunhanlar N, Shpak G, van der Kroeg M, Gouty-Colomer LA, Munshi ST, Lendemeijer B, et al. A simplified protocol for differentiation of electrophysiologically mature neuronal networks from human induced pluripotent stem cells. *Mol Psychiatry*. 2018 May;23(5):1336-44.

RESEARCH ARTICLE

44. Ben Mahmoud M, Rátkai A, Bauer K, Bencsik N, Szücs A, Schlett K, et al. Multifactorial approach is needed to unravel the maturation phases of human neurons derived from induced pluripotent stem cells. *Sci Rep*. 2025 Jan 21;15(1):2627.
45. Rosebrock D, Arora S, Mutukula N, Volkman R, Gralinska E, Balaskas A, et al. Enhanced cortical neural stem cell identity through short SMAD and WNT inhibition in human cerebral organoids facilitates emergence of outer radial glial cells. *Nat Cell Biol*. 2022 June;24(6):981–95.
46. Chen JK, Taipale J, Cooper MK, Beachy PA. Inhibition of Hedgehog signaling by direct binding of cyclopamine to Smoothened. *Genes Dev*. 2002 Nov 1;16(21):2743–8.
47. Cao SY, Hu Y, Chen C, Yuan F, Xu M, Li Q, et al. Enhanced derivation of human pluripotent stem cell-derived cortical glutamatergic neurons by a small molecule. *Sci Rep*. 2017 June 12;7(1):3282.
48. Lendahl U, Zimmerman LB, McKay RDG. CNS stem cells express a new class of intermediate filament protein. *Cell*. 1990 Feb 23;60(4):585–95.
49. Tole S, Goudreau G, Assimacopoulos S, Grove EA. Emx2 Is Required for Growth of the Hippocampus But Not for Hippocampal Field Specification. *J Neurosci*. 2000 Apr 1;20(7):2618–25.
50. Hanashima C, Fernandes M, Hebert JM, Fishell G. The Role of Foxg1 and Dorsal Midline Signaling in the Generation of Cajal-Retzius Subtypes. *J Neurosci*. 2007 Oct 10;27(41):11103–11.
51. Knoth R, Singec I, Ditter M, Pantazis G, Capetian P, Meyer RP, et al. Murine Features of Neurogenesis in the Human Hippocampus across the Lifespan from 0 to 100 Years. *PLOS ONE*. 2010 Jan 29;5(1):e8809.
52. Gantner CW, Hunt CPJ, Niclis JC, Penna V, McDougall SJ, Thompson LH, et al. FGF-MAPK signaling regulates human deep-layer corticogenesis. *Stem Cell Rep*. 2021 Apr 8;16(5):1262–75.
53. Hodge RD, Nelson BR, Kahoud RJ, Yang R, Mussar KE, Reiner SL, et al. Tbr2 Is Essential for Hippocampal Lineage Progression from Neural Stem Cells to Intermediate Progenitors and Neurons. *J Neurosci*. 2012 May 2;32(18):6275–87.
54. Karalay Ö, Doberauer K, Vadodaria KC, Knobloch M, Berti L, Miquelajauregui A, et al. Prospero-related homeobox 1 gene (Prox1) is regulated by canonical Wnt signaling and has a stage-specific role in adult hippocampal neurogenesis. *Proc Natl Acad Sci*. 2011 Apr 5;108(14):5807–12.
55. Nishimura H, Li Y. Human pluripotent stem cell-derived models of the hippocampus. *Int J Biochem Cell Biol*. 2024 Dec;177:106695.
56. Cembrowski MS, Wang L, Sugino K, Shields BC, Spruston N. Hipposeq: a comprehensive RNA-seq database of gene expression in hippocampal principal neurons. Marder E, editor. *eLife*. 2016 Apr 26;5:e14997.
57. Iqbal K, Liu F, Gong CX, Grundke-Iqbal I. Tau in Alzheimer Disease and Related Tauopathies. *Curr Alzheimer Res*. 7(8):656–64.
58. Cid E, Marquez-Galera A, Valero M, Gal B, Medeiros DC, Navarron CM, et al. Sublayer- and cell-type-specific neurodegenerative transcriptional trajectories in hippocampal sclerosis. *Cell Rep* [Internet]. 2021 June 8 [cited 2025 Jan 22];35(10). Available from: [https://www.cell.com/cell-reports/abstract/S2211-1247\(21\)00580-5](https://www.cell.com/cell-reports/abstract/S2211-1247(21)00580-5)
59. Li JT, Xie XM, Yu JY, Sun YX, Liao XM, Wang XX, et al. Suppressed Calbindin Levels in Hippocampal Excitatory Neurons Mediate Stress-Induced Memory Loss. *Cell Rep*. 2017 Oct 24;21(4):891–900.
60. Goffigan-Holmes J, Sanabria D, Diaz J, Flock D, Chavez-Valdez R. Calbindin-1 expression in the hippocampus following neonatal hypoxia-ischemia and therapeutic hypothermia and deficits in spatial memory. *Dev Neurosci*. 2019 Mar 12;1–15.
61. Kohara K, Pignatelli M, Rivest AJ, Jung HY, Kitamura T, Suh J, et al. Cell type-specific genetic and optogenetic tools reveal hippocampal CA2 circuits. *Nat Neurosci*. 2014 Feb;17(2):269–79.
62. Szabadics J, Varga C, Brunner J, Chen K, Soltesz I. Granule Cells in the CA3 Area. *J Neurosci*. 2010 June 16;30(24):8296–307.
63. Huang X, Chen Q, Luo W, Pakvasa M, Zhang Y, Zheng L, et al. SATB2: A versatile transcriptional regulator of craniofacial and skeleton development, neurogenesis and tumorigenesis, and its applications in regenerative medicine. *Genes Dis*. 2022 Jan 1;9(1):95–107.
64. Otani T, Marchetto MC, Gage FH, Simons BD, Livesey FJ. 2D and 3D Stem Cell Models of Primate Cortical Development Identify Species-Specific Differences in Progenitor Behavior Contributing to Brain Size. *Cell Stem Cell*. 2016 Apr 7;18(4):467–80.
65. Hergenreder E, Minotti AP, Zorina Y, Oberst P, Zhao Z, Munguba H, et al. Combined small-molecule treatment accelerates maturation of human pluripotent stem cell-derived neurons. *Nat*

RESEARCH ARTICLE

- Biotechnol [Internet]. 2024 Jan 2 [cited 2024 Jan 5]; Available from: <https://www.nature.com/articles/s41587-023-02031-z>
66. Savage JT, Ramirez JJ, Risher WC, Wang Y, Irala D, Eroglu C. SynBot is an open-source image analysis software for automated quantification of synapses. *Cell Rep Methods* [Internet]. 2024 Sept 16 [cited 2026 Jan 16];4(9). Available from: [https://www.cell.com/cell-reports-methods/abstract/S2667-2375\(24\)00239-X](https://www.cell.com/cell-reports-methods/abstract/S2667-2375(24)00239-X)
67. Yakoub AM. Cerebral organoids exhibit mature neurons and astrocytes and recapitulate electrophysiological activity of the human brain. *Neural Regen Res*. 2019 May;14(5):757–61.
68. Amin ND, Kelley KW, Kaganovsky K, Onesto M, Hao J, Miura Y, et al. Generating human neural diversity with a multiplexed morphogen screen in organoids. *Cell Stem Cell*. 2024 Dec 5;31(12):1831-1846.e9.
69. Kagan BJ. The CL1 as a platform technology for leveraging biological neural system functions. *Nat Rev Bioeng*. 2025;
70. Kagan BJ, Habibollahi F, Watmuff B, Azadi A, Doensen F, Loeffler A, et al. Harnessing intelligence from brain cells in vitro. *The Neuroscientist*. 2025;
71. Alam El Din DM, Moenkemoeller L, Loeffler A, Habibollahi F, Schenkman J, Mitra A, et al. Human neural organoid microphysiological systems show the building blocks necessary for basic learning and memory. *Commun Biol*. 2025 Aug 16;8(1):1237.
72. Habibollahi F, Kagan BJ, Burkitt AN, French C. Critical dynamics arise during structured information presentation within embodied in vitro neuronal networks. *Nat Commun*. 2023 Aug 30;14(1):5287.
73. Kagan BJ. Two roads diverged: Pathways toward harnessing intelligence in neural cell cultures. *Cell Biomater*. 2025 Sept;1(8):100156.
74. Watmuff B, Habibollahi F, Desouza C, Khajehnejad M, Loeffler A, Baranes K, et al. Drug treatment alters performance in a neural microphysiological system of information processing. *Commun Biol*. 2025 June 17;2025.01.27.635163.
75. Shi Y, Kirwan P, Livesey FJ. Directed differentiation of human pluripotent stem cells to cerebral cortex neurons and neural networks. *Nat Protoc*. 2012 Oct;7(10):1836–46.
76. Watmuff B, Habibollahi F, Desouza C, Khajehnejad M, Loeffler A, Baranes K, et al. Drug treatment alters performance in a neural microphysiological system of information processing. *Commun Biol*. 2025 June 17;8(1):916.
77. Abu-Bonsrah KD, Zhang D, Bjorksten AR, Dottori M, Newgreen DF. Generation of Adrenal Chromaffin-like Cells from Human Pluripotent Stem Cells. *Stem Cell Rep*. 2018 Jan 9;10(1):134–50.
78. Development of Functional Thyroid C Cell-like Cells from Human Pluripotent Cells in 2D and in 3D Scaffolds [Internet]. [cited 2026 Jan 21]. Available from: <https://www.mdpi.com/2073-4409/10/11/2897>
79. Herculano-Houzel S. The glia/neuron ratio: How it varies uniformly across brain structures and species and what that means for brain physiology and evolution. *Glia*. 2014;62(9):1377–91.
80. Bass NH, Hess HH, Pope A, Thalheimer C. Quantitative cytoarchitectonic distribution of neurons, glia, and DNA in rat cerebral cortex. *J Comp Neurol*. 1971;143(4):481–90.
81. Tanveer MS, Patel D, Schweiger HE, Abu-Bonsrah KD, Watmuff B, Azadi A, et al. Starting a synthetic biological intelligence lab from scratch. *Patterns* [Internet]. 2025 May 9 [cited 2025 Oct 29];6(5). Available from: [https://www.cell.com/patterns/abstract/S2666-3899\(25\)00080-7](https://www.cell.com/patterns/abstract/S2666-3899(25)00080-7)
82. Beggs JM, Timme N. Being Critical of Criticality in the Brain. *Front Physiol* [Internet]. 2012 [cited 2024 May 16];3. Available from: <http://journal.frontiersin.org/article/10.3389/fphys.2012.00163/abstract>
83. Yeomans J. Statistical mechanics of phase transitions. Clarendon Press; 1992.
84. Nishimori H, Ortiz G. Elements of phase transitions and critical phenomena. Oxford university press; 2011.
85. Klaus A, Yu S, Plenz D. Statistical Analyses Support Power Law Distributions Found in Neuronal Avalanches. Zochowski M, editor. *PLoS ONE*. 2011 May 26;6(5):e19779.
86. Granger CWJ. Investigating Causal Relations by Econometric Models and Cross-spectral Methods. *Econometrica*. 1969;37(3):424–38.
87. Watts DJ, Strogatz SH. Collective dynamics of ‘small-world’ networks. *Nature*. 1998 June;393(6684):440–2.
88. Onnela JP, Saramäki J, Kertész J, Kaski K. Intensity and coherence of motifs in weighted complex networks. *Phys Rev E*. 2005 June 13;71(6):065103.